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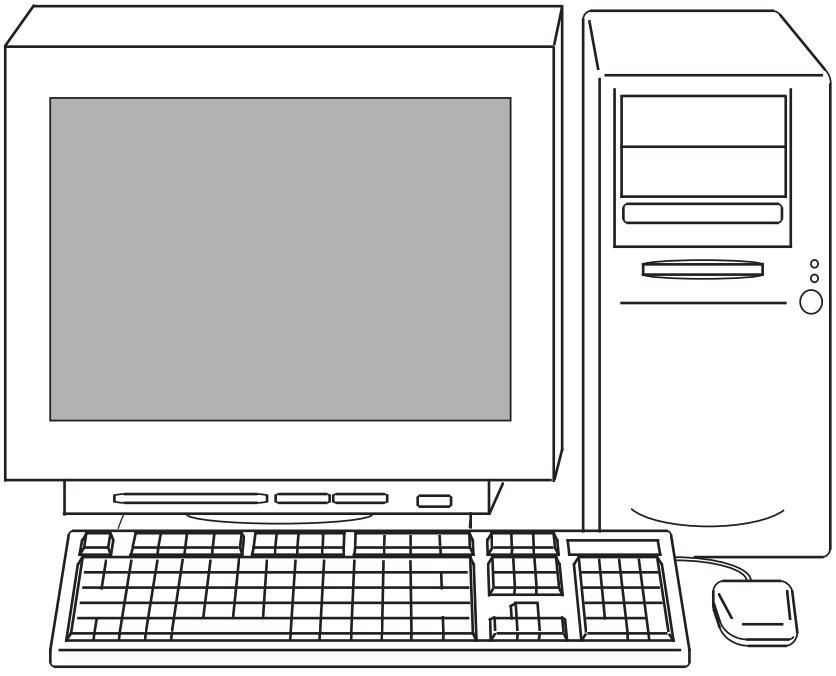
Operating Instructions

Enterprise Link

MES for Electronics Assembly

PanaCIM-EE

Model No. **NM-EJS3B**



Thank you for purchasing the “Panasonic product.”
- Read the operating instructions carefully for proper and safe usage.
- Before placing the machine in service, be sure to read “Safety Precautions.”
- Store the operating instructions carefully.

N7201A522E11 (V08.08)
201801

Original Instructions

For using this system:
This system is designed with a purpose of visualization of work productivity, more efficiency and quality warranty in electronic component mounting process. Use this system solely for the purpose.

Do not use this system for other purposes.

Operating machines:
Any person who operates the machine should receive a special training.

Any operator who has not received the training is not allowed to operate.

For training, contact Panasonic Smart Factory Solutions Co., Ltd., Sales company or Sales agent.

⚠ Warning

Do not operate the machine without receiving special training



Risk of injury due to improper operation.

Prohibited

●About training, conduct in your company or contact us.

For machine training in Europe, you may contact Panasonic Industry Europe:

Panasonic Industry Europe GmbH FA Division
Robert-Koch-Strasse 100, 85521 Ottobrunn Germany
Phone number: 0049-(0)89-453541192

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About the Instruction

■ Operating instruction contents

The following operating instructions are included in the DVD-ROM attached to this machine. The instructions are divided according to work or function. Use one that is appropriate for the purchased module (application) or your purpose.

∗ BM series machine is included in the explanation of this manual, but its functions are not supported.

Type	Description
<General>	
Introduction	Overview of machine and basic setting/operation/analysis/utility.
Basic Operation (PanaCIM Web Application)	PanaCIM Web application's basic operation.
<Modules>	
Production Analysis	Collect and analyze machine's running information and display its statistics on real time basis to support productivity improvements (recognition/pickup error rate improvement, machine trouble stop time reduction) for mounting line.
Traceability	Combine PCB information read by a barcode reader on machine or conveyor and production record or material/part information of machine to manage history of PCBs for quality control and market support.
Material Verification	Verify material/part information and recipes while setting up material/parts in production setup/model change/splicing/part exhaust to prevent wrong material/part set up.
Product Changeover and Control	Auto download recipe and adjust lane width based on PCB ID for quicker model change.
Production Monitoring	Retrieve production information and operation information on real time basis and display line status on the monitor and PDA to provide operators with instructions for events such as error or part exhaust. This is to reduce mount line stop time or to prepare in advance for better productivity.
Enterprise Link	A function to output the data collected from machines during production (material verification data, trace data, operation performance data) in the standard format so external applications can reuse the data. Part information can be loaded also in the standard format.

Type	Description
Material Control	A function to provide the location and the number of parts or feeders stocked in the production or preparation area. MSD function controls MSD parts' expired date.
Component Supply Navigator	The Component Supply Navigator optimizes the supply sequence to provide a shortest route of an operator without component shortage. It can prevent short time breakdown in the production line occurred due to component shortage and contribute to manpower saving.
Maintenance	For the purpose of reducing time spent on maintenance planning / control operations, asset (machines, heads and feeders) profiles can be controlled on the floor, or the maintenance history of each asset can be recorded.

License CD names for modules are as follows.

Module Name	License CD Name
Production Analysis	Production Analysis
Traceability	Traceability
Material Verification	Material Verification
Product Changeover and Control	Product Changeover and Control
Production Monitoring	Production Monitoring & Dispatch
Enterprise Link	Enterprise Link
Material Control Base	Material Control Base
Material Control Allocation	Material Control Allocation
Material Control MSD	Material Control MSD
Component Supply Navigator	Component Supply Navigator
Maintenance	Maintenance

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1. OVERVIEW

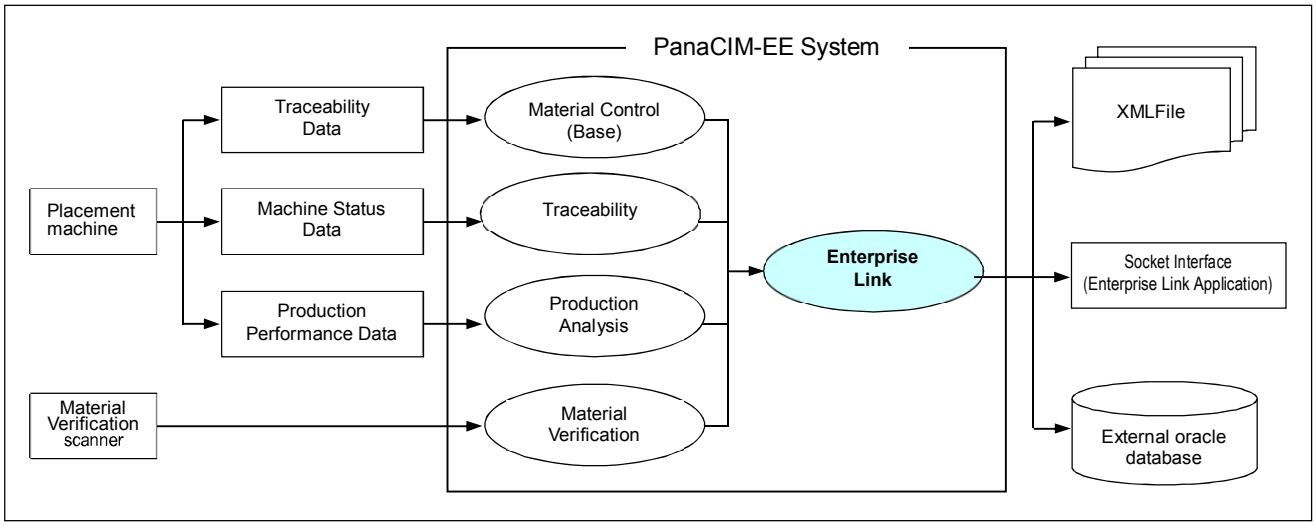
This chapter explains the overview of PanaCIM-EE / Enterprise Link.

1.1 Overview

PanaCIM-EE(PanaCIM Enterprise Edition: hereinafter referred to as “PanaCIM”) Enterprise Link module allows the users to export the data collected from the machines during production (material verification data, traceability data, and production performance data) in standard format so that such data may be reused by external application programs.
It also enables the component information to be imported in standard format.

■ Main functions

1. Exporting data as XML files through socket interface or external database interface.
   * The data that can be exported are panel traceability data, material verification data, machine status data, and production data.
2. Acting as a central repository that ensures proper data transfer and handles data in the event of errors for highly reliable asynchronous LAN communication.



■ Enterprise Link module functions

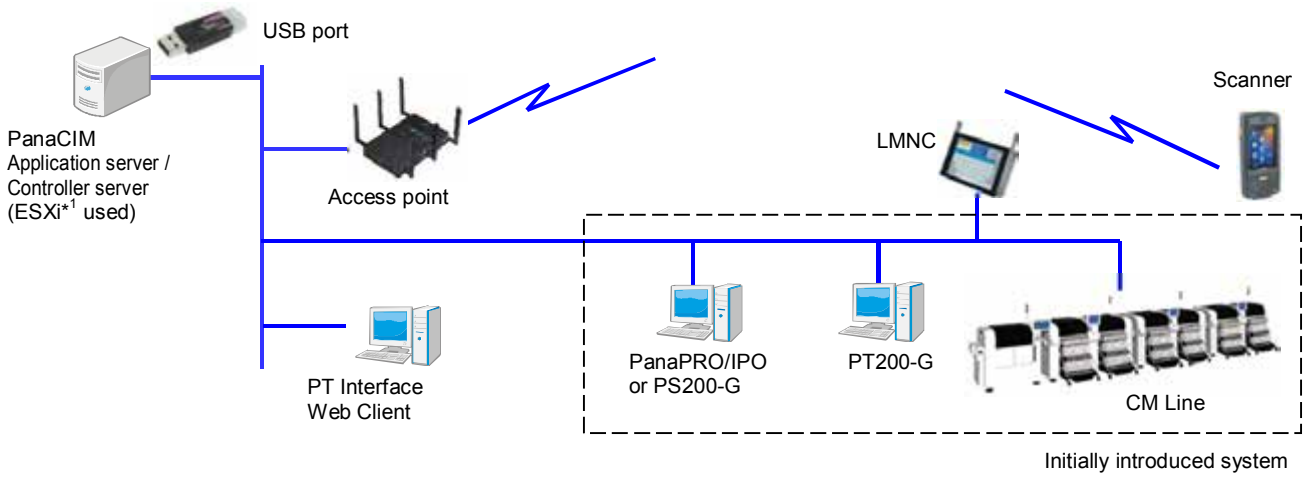
Main function	Detailed functions		Description
	Large category	Small category	
Export	XML		XML Interface
	Socket		Socket Interface
	Database		Database Interface

1.2 System Configuration

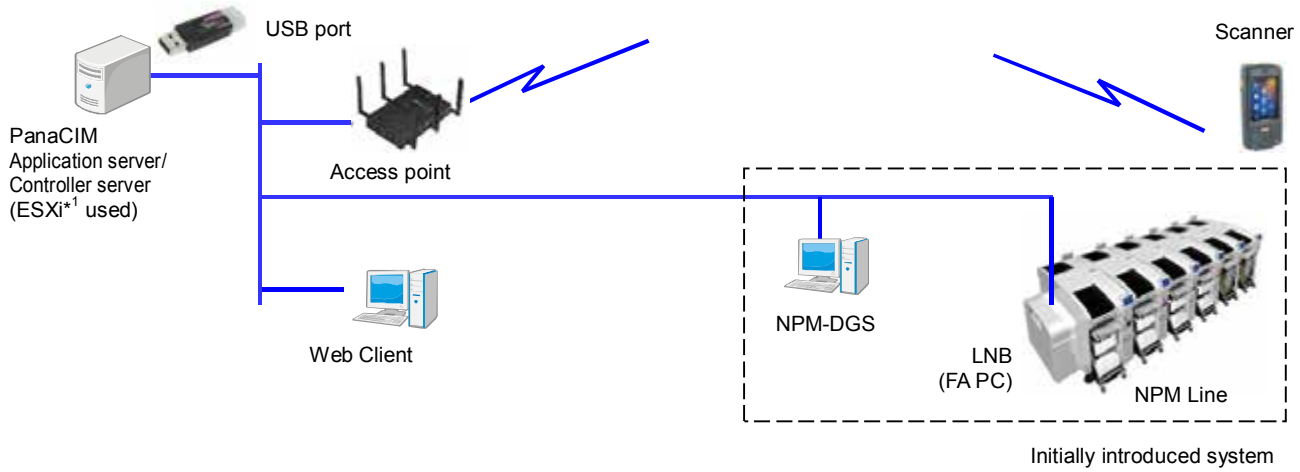
This section shows the schematics of systems with PanaCIM for small-scale and large-scale production.

■ System with PanaCIM (for small-scale production)

CM Line



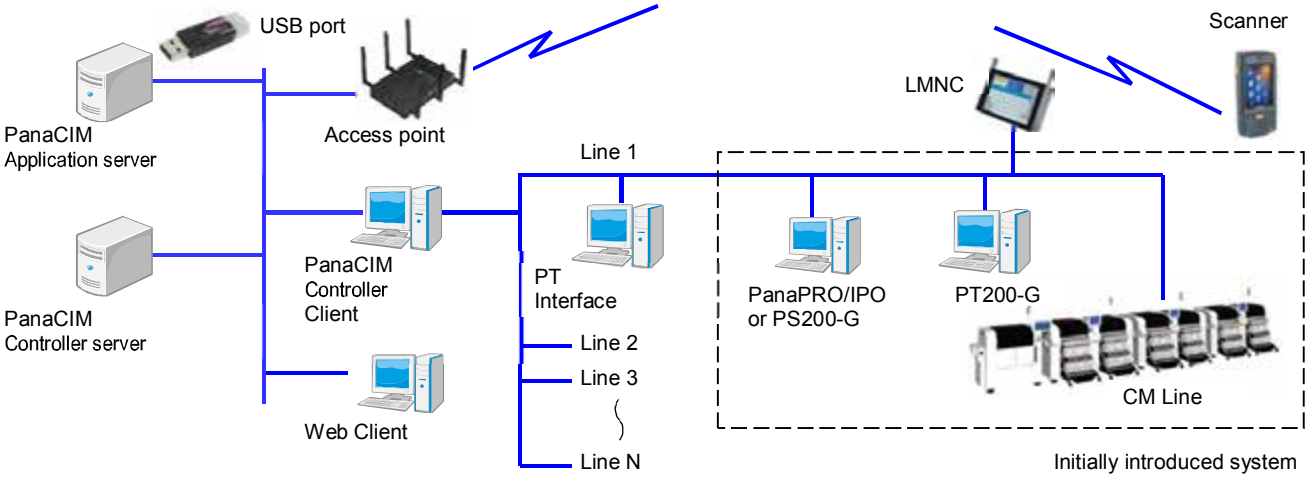
NPM Line*2



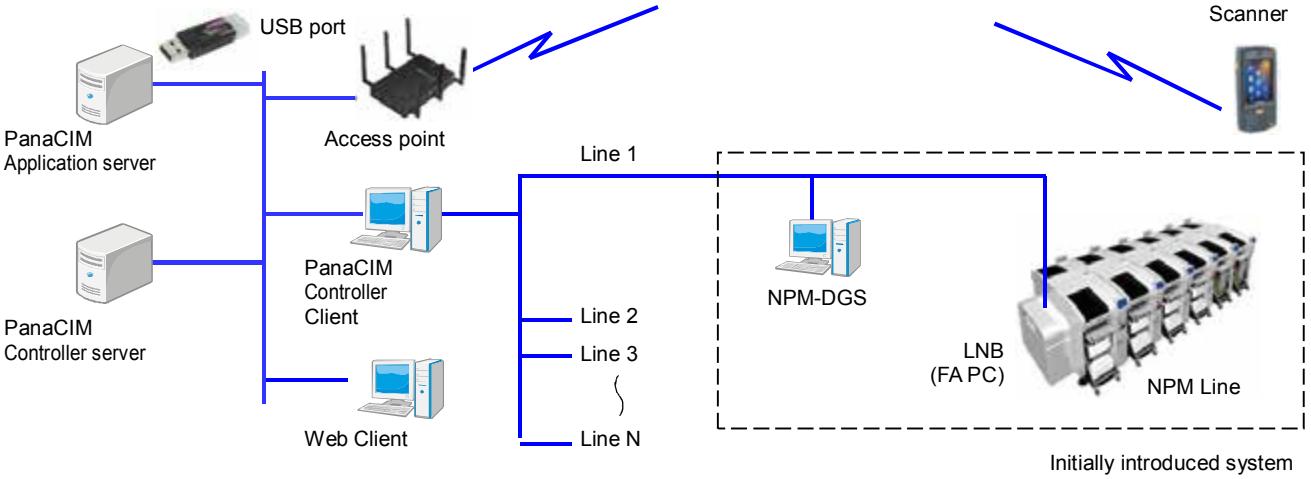
*1. ESXi: A virtualization tool that allows multiple applications to run on a single server.
*2. NPM line can also include AM100 as NPM-related machine.

■ System with PanaCIM (for large-scale production)

CM Line



NPM Line*1



*1. NPM line can also include AM100 as NPM-related machine.

1.3 Functional Overview

■ List of Enterprise Link module functions

Main functions	Detailed functions		Description
	Large category	Small category	
Export	XML		XML Interface
	Socket		Socket Interface
	Database		Database Interface

Main functions	Detailed functions		Description	Required licenses			
	Output format	Item		Material Verifi-cation	Material Control	Produc-tion Analysis	Trace-ability
Data Export	XMLFile Export	Traceability data	Exports the traceability data from the PanaCIM database in the XML format.	○	—	—	○
		Machine status data	Exports the machine status data from the PanaCIM database in the XML format.	—	—	○	—
		Scrap data	Exports the scrap data from the PanaCIM database in the XML format.	—	—	○	—
		Product data	Exports the product data from the PanaCIM database in the XML format.	○	—	○	—
		Material verification data	Exports the material verification data from the PanaCIM database in the XML format.	○	—	—	—
	Database Export	Material verification data	Exports the material verification data from the PanaCIM database to the external database.	○	—	—	—
	Socket Channel	Material verification data	Exports the material verification data from the PanaCIM database to the external application.	○	—	—	—
Data Import	CSV File Import	Restricted/Invalid components	Imports the data of the restricted/ invalid components in the CSV format.	○	○	○	

2. SETTINGS

This chapter explains a setup of PanaCIM-EE / Enterprise Link.

2.1 Overview

2.1.1 Typical Settings

The Typical settings for Enterprise Link are as follows:

Enterprise Link Settings			
		Name of the control	Value (Typical setting)
Module Configuration			
	Enterprise Link		
		General Settings	
		Enable Debug Logs	**unchecked**
		Delete files older than (Days)	**1**
		Channel Information Setup->Channel Selections	
		Database	**unchecked**
		CAMX	**unchecked**
		Channel Information Setup->Database Tab (if checked for first time)	MS SQL	**unselected**
		Oracle	**unselected**
		Channel Information Setup->Socket Tab	Add Button	**Enabled**
		Edit Button	**Disabled**
		Delete Button	**Disabled**
		Channel Information Setup->Web Service Tab	Add Button	**Enabled**
		Edit Button	**Disabled**
		Delete Button	**Disabled**

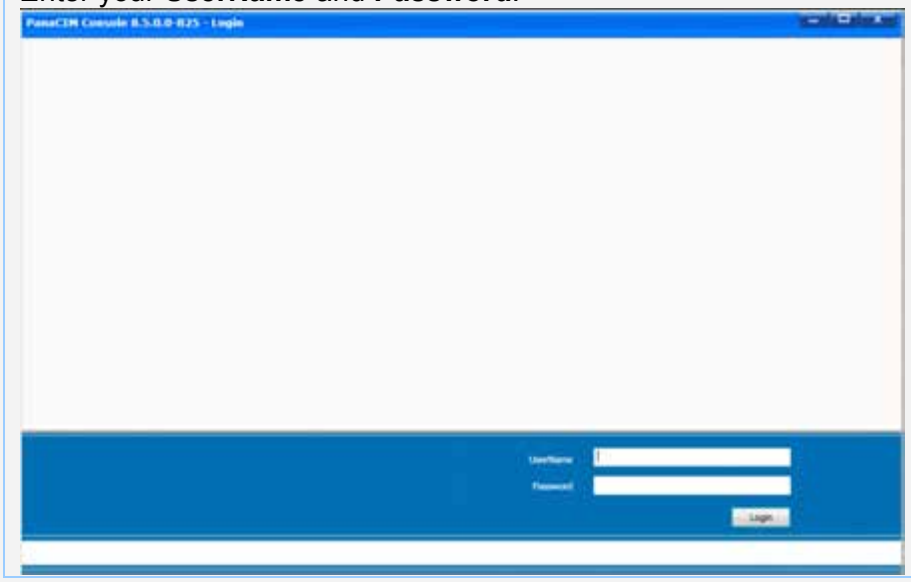
2.2 Settings

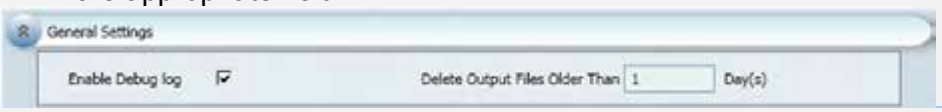
This section describes the procedure for setting the PanaCIM Console for Enterprise Link and Service Controller.

Application	Description
PanaCIM Console for Enterprise Link	PanaCIM Console has been installed on Windows. The following settings are available using this console: • Database setting • Data import setting (including MCID) • Data export setting
PanaCIM Service Controller	Contains a number of integrated services provided by the PanaCIM application server. The Enterprise Link can be started, stopped or restarted via this controller.

2.2.1 General Settings

Refer to the action table below to configure the general settings using the PanaCIM Console.

Step	Action
1	Select **Start > Programs > Panasonic > PanaCIM > PanaCIM Services > PanaCIM Console**.
2	Enter your **UserName** and **Password**. 
3	Select the **Module Configuration** link and then expand the **Enterprise Link** Configuration section to select **General Settings**.

	
4	Expand **General Settings**. a. Select the **Enable Debug log** checkbox to create a log file of errors and system messages. b. Enter the number of day(s) (1-999) before the system **deletes old files** in the appropriate field. 
5	Click **Save** and restart Enterprise Link, or continue on to Channel Information Setup in the next section.

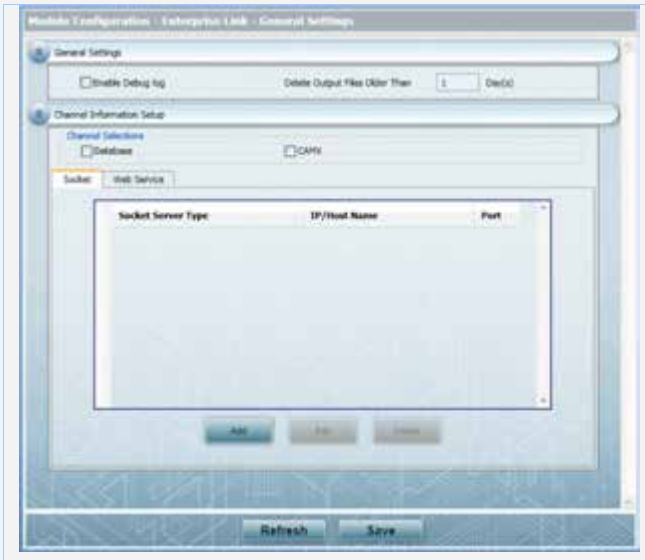
2.2.2 Channel Information

Unless otherwise specified, data are exported in the XML format.

Database
CAMX: CAM using XML (or CAMX) is used as a data conversion tool for the SMT lines and applications on the shopfloor.
Socket: Used as a two-way communication link between two programs running on the Internet

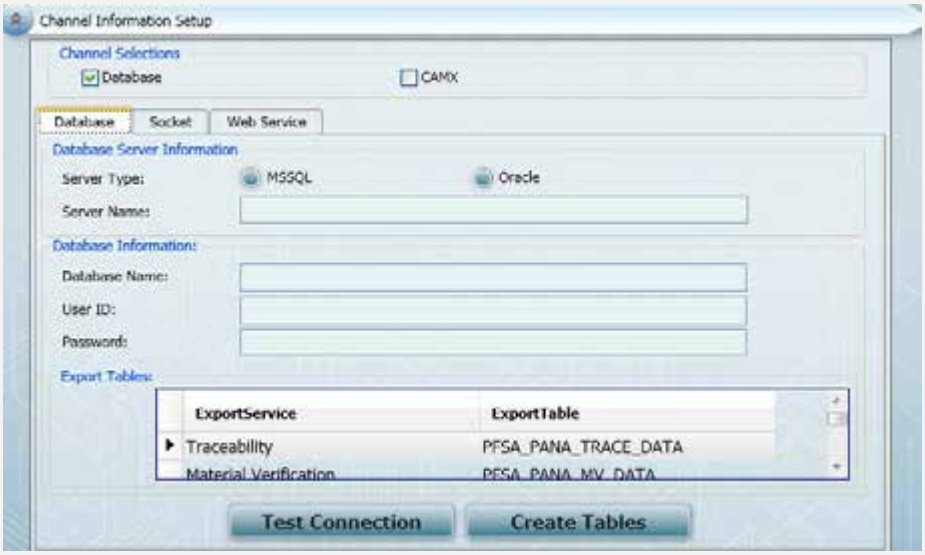
The following describes how the traceability data is exported to the database using these three channels.

Step	Action
1	Select the **Module Configuration** link and then expand the **Enterprise Link** Configuration section to select **General Settings**. 
2	Expand **General Settings**. Select how information should be exported; by Database or CAMX. **Socket**: Enter the IP Address / Host name and port number of the server to which data needs to be exported.



Database: Select the Database checkbox to export information through the database.

a. Select **MSSQL** from the Server Type field.
b. Enter the **SQL Server Name** where the data is being exported.
c. Enter the **Database Name** where the data is being exported.
d. Enter the **User ID**.
e. The default table (Export Table) will be created in the Database after you have entered the Database Server information.
f. Click **Test Connection** to test the connection to the database, and make sure no error occurs.



CAMX: Select the CAMX checkbox to export information by CAMX.

a. Enter the **Name**.
b. Enter the **Domain Name**.
c. Enter the **URL** address.

	
3	Click **Save**.
4	You need to restart Enterprise Link Service to make the configuration effective. a. Select **Programs > Panasonic > PanaCIM > PanaCIM Services > ServiceController**. b. Select **Enterprise Link** and click **Restart**. 

3. FUNCTION - IMPORT

This chapter explains function-import.

3.1 Overview

Use the Enterprise Link Configuration Manager within the PanaCIM Console to activate import settings. Once configured, you will be able to enable/disable the importing of data, as well as update the directory where files will be located for importing. The Enterprise Link also allows configuration for and supports import from external 3rd party client systems. Enterprise Link will monitor a user defined directory for CSV files and when a file is detected, Enterprise Link will read and import the file and its contents.

Once data has been imported, checks for these components will be completed during the following timeframes:

- During part scan operations
- Production start
- When an update or change is made to the component list

The import function includes:

- Invalid/Restricted Component
- MCID
- Material Change
- Part Library

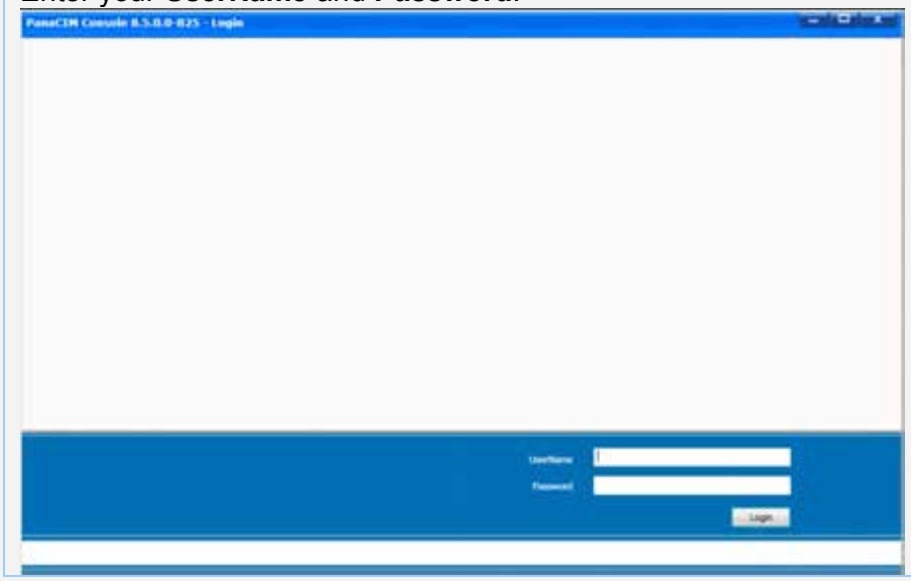
3.2 Settings

3.2.1 Invalid/Restricted Component

■ Invalid/Restricted Component Import Settings

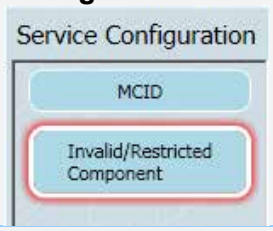
Refer to the action table below to configure import settings for invalid/restricted components.

Note: Activating Imports of Invalid/Restricted components via Enterprise Link will disable the editing of Invalid/Restricted components via Inventory Manager. Using the Enterprise Link import expedites configuration of Invalid/Restricted Components by using a .CSV file. To import these components using Inventory Manager, please reference the Material Verification documentation.

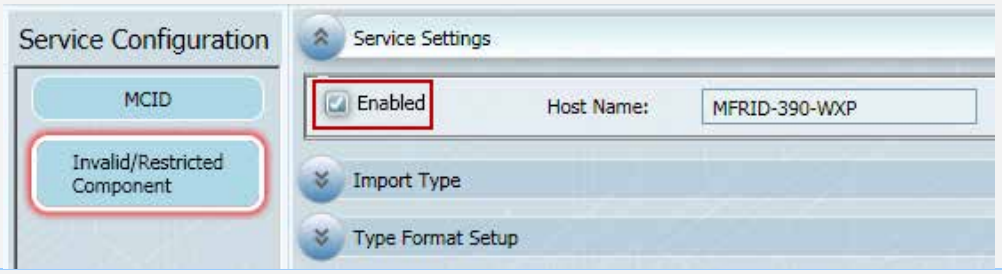
Step	Action
1	Select **Start > Programs > Panasonic > PanaCIM > PanaCIM Services > PanaCIM Console**.
2	Enter your **UserName** and **Password**. 
3	Select the **Module Configuration** link and then expand the **Enterprise Link** Configuration section to select **Import Settings**.



4 Click the **Invalid/Restricted Component** link under the **Service Configuration** section.

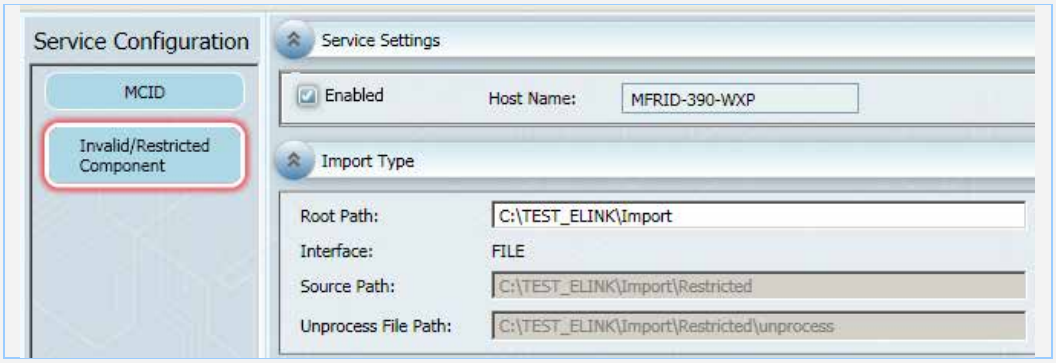


5 Expand **Service Settings** and select the **Enabled** checkbox to enable these settings on the Host PC. The name of the Host PC will appear in the Host Name field.



6 Expand **Import Type**.

- The **Root Path** field is editable and can be modified if desired.
- The Source Path field provides the name of the folder path where the .CSV files are stored and cannot be edited.



7 Expand **Type Format Setup**. This section is where you will define how the .CSV files are imported.

a. Files must be comma separated, so the **Delimiter Character** should remain as ‘**,**’.

b. Add a number of lines to be skipped in the .CSV files in the **Skip First N Lines** field. If you generally use the first couple of rows or so to add comments into the file, these are the rows that should be skipped when importing data from the file.

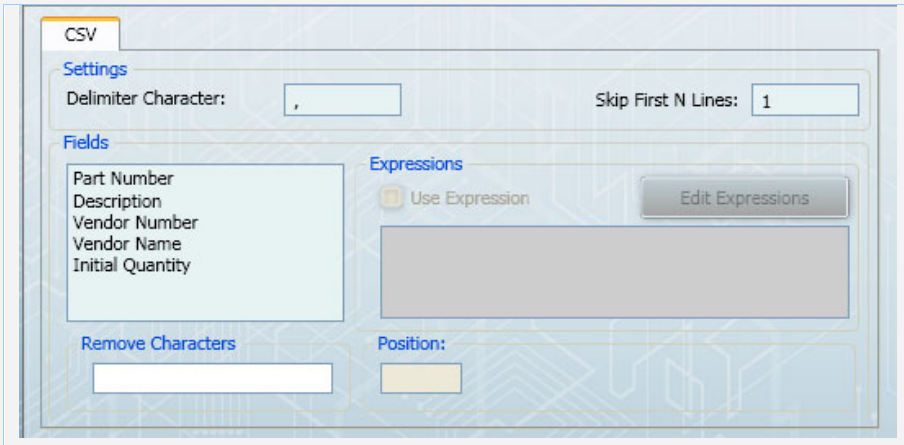
c. Select how to parse the file in the **Fields** selection box. Select each Field one at a time to input specific information related to that field. Reference the below table for field descriptions.

Notes:

- As you click on each field, the Position will appear in the provided field. Change the field position as desired.
- Remove any characters from the Part Number expression; add it into the Remove Characters field.
- To edit an expression, click Edit Expressions to initiate a dialog box where you can add, edit, or delete expressions.

Field	Description
PRODUCT_NAME	This must contain one of the following: PanaPRO/IPO Mix_Name, PT200 MCFile Name, or DGS Lot Name.
PART_NO	Part number that is to be restricted.
VENDOR_NO	Vendor number that will be associated with the part number further defining its restricted attribute status.
LOT_NO	The lot number that will be associated with the part number further defining its restricted attribute status.
USER_DATA	The user defined data that will be associated with the part number further defining its restricted attribute status.

Note: The maximum character length for each field is 30 characters.



8 Click **Save**.

9 You need to restart Enterprise Link Service to make the configuration effective.

a. Select **Service > Service Controller**.

b. Select **Enterprise Link** and click **Restart**.

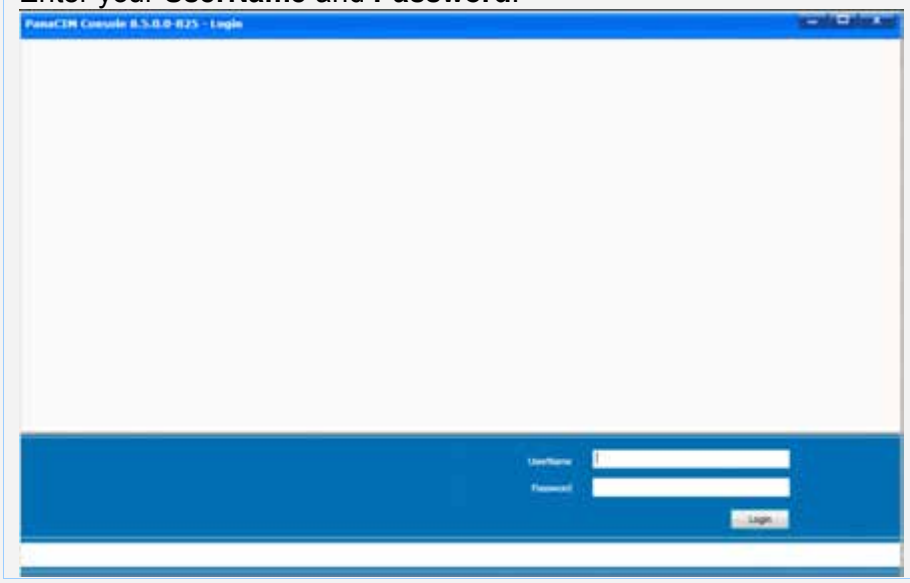


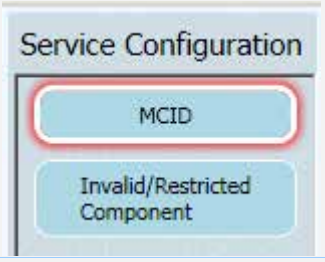
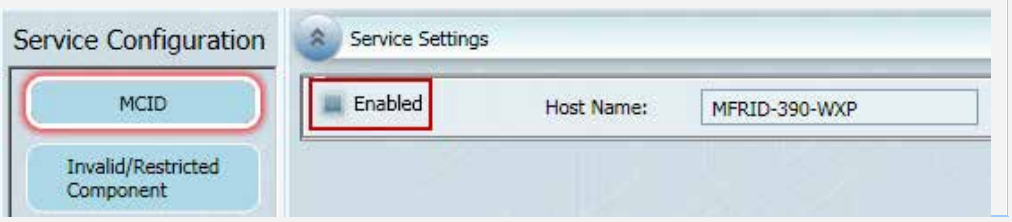
3.2.2 MCID

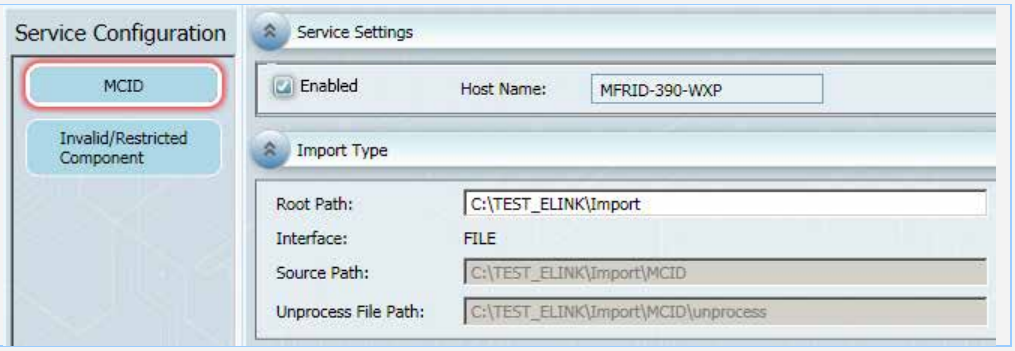
■ MCID Import Settings

Refer to the action table below to import MCID.
(MCID refers to the user-defined material ID.)

Note: These settings are optional and only apply if using MCID.

Step	Action
1	Select **Start > Programs > Panasonic > PanaCIM > PanaCIM Services > PanaCIM Console**.
2	Enter your **UserName** and **Password**. 
3	Select the **Module Configuration** link and then expand the **Enterprise Link** Configuration section to select **Import Settings**.

	
4	Click the **MCID** link under the Service Configuration section. 
5	Expand **Service Settings** and select the **Enabled** checkbox to enable these settings on the Host PC. 
6	Expand **Import Type**. • The **Root Path** field is editable and can be modified if desired. • The **Source Path** field provides the name of the folder path where the CSV files are stored and cannot be edited.



7 Expand **Type Format Setup**.

The **CSV** tab is selected by default. Use this tab to configure the MCID file format, and MCID/Part_No Expressions:

a. Files must be comma separated, so the **Delimiter Character** should remain as '**,**'.

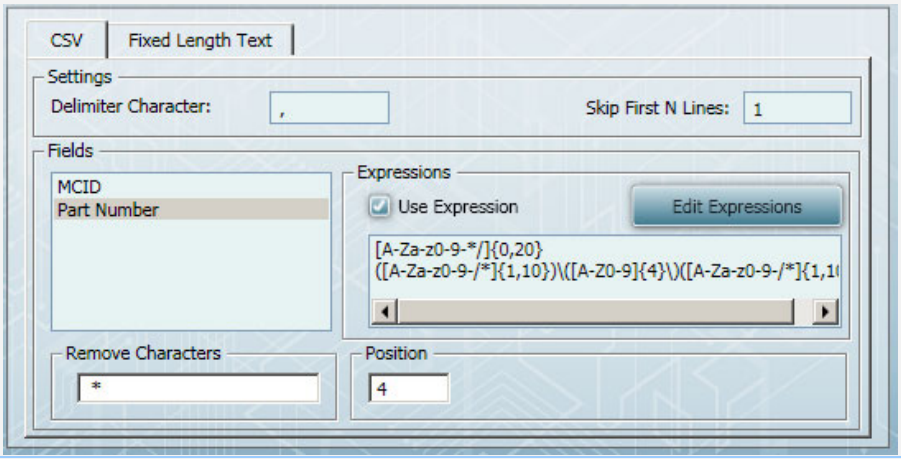
b. Add a number of lines to be skipped in the .CSV files in the **Skip First N Lines** field. If you generally use the first couple of rows or so to add comments into the file, these are the rows that should be skipped when importing data from the file.

c. Select how to parse the file in the **Fields** selection box.

- If you select MCID, the Expressions field will remain blank. MCID is processed as it is defined in the file.
- Expressions are pre-defined for Part Number, but can be modified by selecting the **Edit Expressions** button. A dialog box is initiated where you can add, edit, or delete an expression.

Notes:

- As you click on each field, the Position will appear in the provided field. Change the field position as desired.
- Remove any characters from the Part Number expression; add it into the Remove Characters field.



8 Click **Save**.

9 Click the **Fixed Length Text** tab.

a. Define settings for the file in the **Settings** section.

- Define the minimum length of the file (or row length) by completing the Minimum Length field.
- Skip First N Lines means to exclude a certain number of lines that start from the top of the file.

b. Select either MCID or Part Number from the **Fields** section.

- If you select MCID, the Expressions field will remain blank. MCID is

	processed as it is defined in the file. • Expressions are pre-defined for Part Number, but can be modified by selecting the **Edit Expressions** button. A dialog box is initiated where you may add, edit, or delete and expression. c. Enter **Start** and **End** Positions for the MCID/Part Number field in a file. d. Enter any character(s) to be excluded in the **Remove Characters** field.
10	Click **Save**.

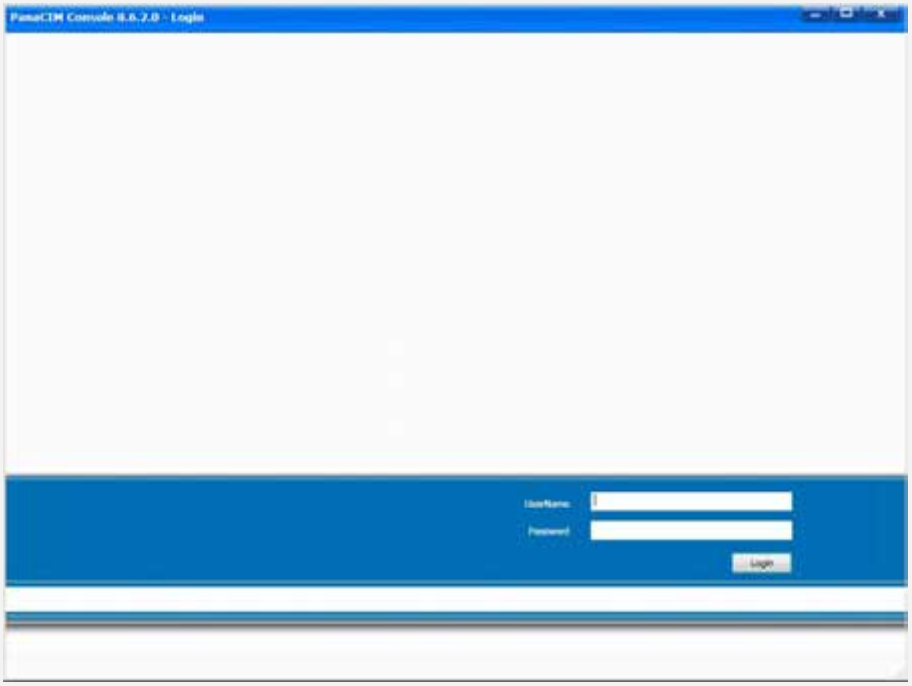
3.2.3 Material Change

■ Material Change Import Settings

This function permits the quantities of materials controlled by the user to be changed at once through the files.
The result of changes can be confirmed in the import log reports.

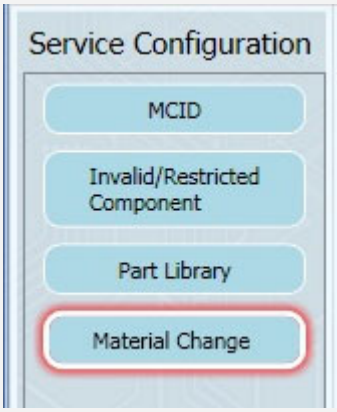
Follow the steps below to import the material change file.
Note: This function is available when the following conditions are met.

- The verification mode is either MC or MV+MatID.
- Areas with “Default for Auto Registration” enabled exist in the route.
- Materials to be changed are checked in (exist) within the above-mentioned area.

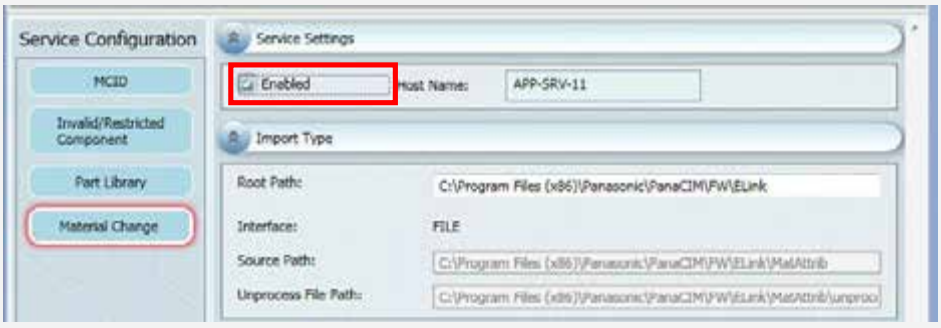
Step	Action
1	Select **Start > Programs > Panasonic > PanaCIM > PanaCIM Services > PanaCIM Console** to launch the PanaCIM Console.
2	Enter your **UserName** and **Password**. 
3	Select the **Module Configuration** link and then expand the **Enterprise Link** Configuration section to select **Import Settings**.



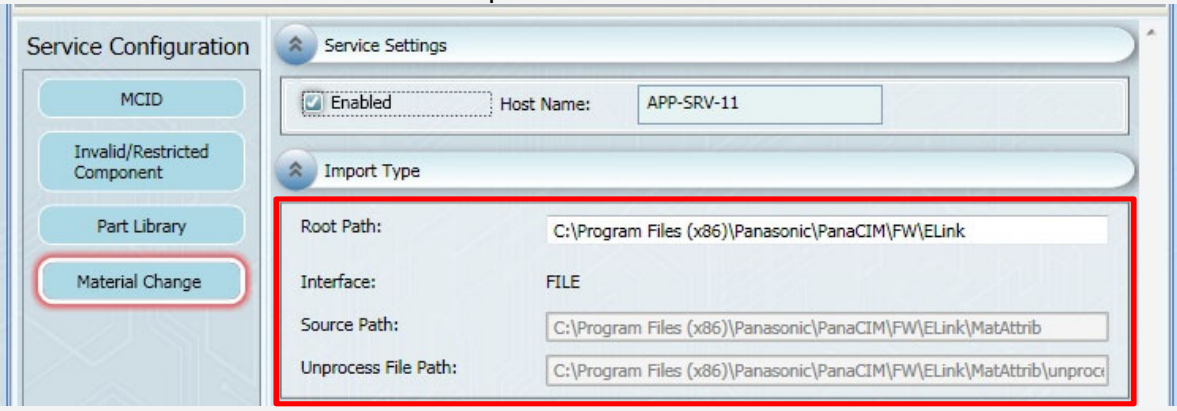
4 Click the **Material Change** link under the **Service Configuration** section.



5 Select the **Enabled** checkbox under **Service Settings** to enable these settings on the Host PC.



6 Expand **Import Type**.

	• Edit the **Root Path** to change the location of files. • The **Source Path** field shows the path to the folder in which files are located. 
7	Click **Save**.

3.3 Operation

3.3.1 Invalid/Restricted Component

Refer to 3.2.1 for details.

3.3.2 MCID

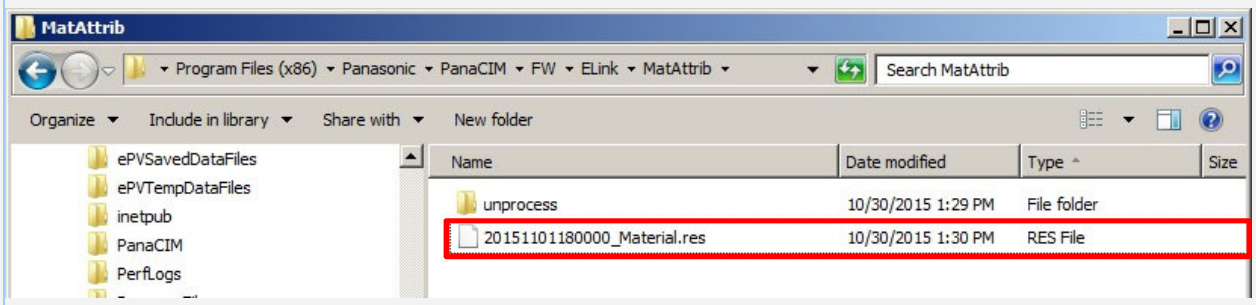
Refer to 3.2.2 for details.

3.3.3 Material Change

■ Importing Material Change Files

Take the steps below to import the material change file.
Note: This function is available when the following conditions are met.

- The verification mode is either MC or MV+MatID.
- Areas with “Default for Auto Registration” enabled exist in the route.
- Materials to be changed are checked in (exist) within the above-mentioned area.

Step	Action
1	Create a file. The file format is made up as follow: File name：XXXX.req * XXXX can be any alphanumeric string. MaterialID:材料 ID,Quantity:数量,Command:Change MaterialID:材料 ID,Quantity:数量,Command:Change MaterialID:材料 ID,Quantity:数量,Command:Change : :
2	Locate the created file in the specified path. * For the file path, refer to step 6 in “3.2.3 Material Change”.
3	The located file will be imported to the PanaCIM. * This file will be removed automatically after imported to the PanaCIM.
4	The result file will be created in the same folder. * This file will not be removed automatically.  File name：XXXX.res * The file will be named the same as the one in step 1 except for the extension (.res). MaterialID:材料 ID,Result:結果 ID MaterialID:材料 ID,Result:結果 ID MaterialID:材料 ID,Result:結果 ID : :

5 The quantities of components will be updated.
* The table below summarizes the meaning of each Result code:

Result code	Description
0	Successfully registered
- 1	Storage unit not registered. * Move (check-in) the materials to the storage unit and try again.
- 2	Unregistered material * Register (check-in) the material to the storage unit and try again.
- 3	Database error * Make sure the database server has been properly connected.
- 4	Quantity exceeds the limit allowed by PanaCIM * Component quantity must be less than 65535.
- 5	Material already reserved. * Cancel the reservation and try again.
- 6	File format error * Make sure the component change file is in correct format.
- 7	An attempt was made to update the material already used up. * Materials cannot be updated if the component quantity is already 0.
- 8	An attempt was made to update the already spliced reel. * Unsplice the reel (cancel splicing) and try again.
- 9	An attempt was made to update to a value more than the initial quantity. * Values more than the initial value cannot be set.
-10	An attempt was made to update the material with ID exceeding 30 digits. * Quantity cannot be changed if the Material ID exceeds 30 digits.
-11	An attempt was made to update to a value less than 0. * Values less than 0 cannot be set.
-12	An attempt was made to update the quantity with the current value. * Make sure the new quantity is correct.

4. FUNCTION - EXPORT

This chapter explains function-export.

4.1 Overview

This section describes how to configure export settings for various PanaCIM Modules. Different export formats are available (XML, Database, Socket, CAMX), depending on the data type being exported and the Machine (Traditional (Panasert), Modular I (CM/DT), Modular II (NPM, including AM100)) used. The table below summarizes the options.

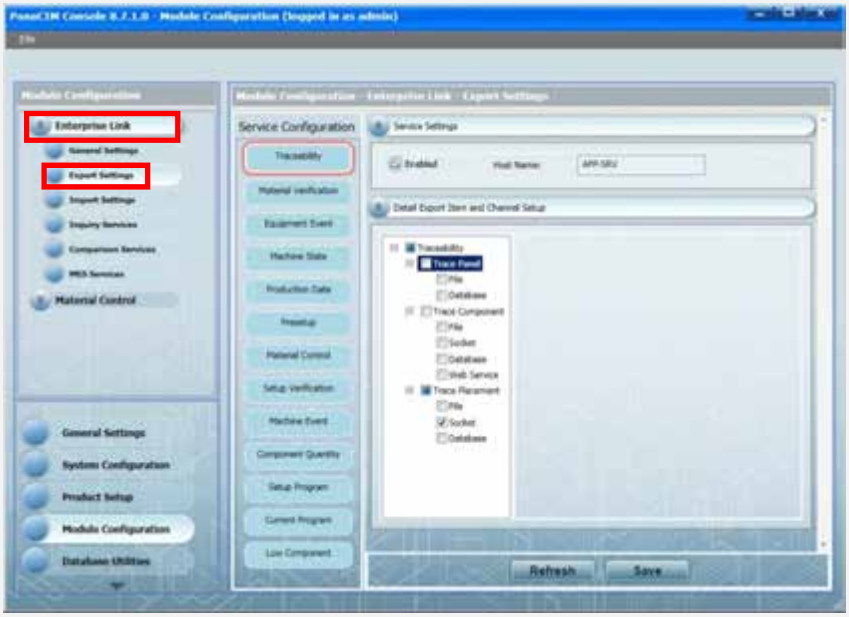
Export Type	Machine Category			Export Channel			
	Trad	Mod I	Mod II	Trad	Mod I	Socket	Trad
Traceability/Panel							
Traceability/ Component	✓	✓	✓	✓	✓	✓	
Traceability/ Placement			✓	✓		✓	
Material Verification	✓	✓	✓	✓			
Machine State	✓	✓	✓	✓			
Equipment Event	✓	✓	✓				✓
Production Data/Scrap Data		✓	✓	✓		✓	
Production Data/Production PCB Data		✓	✓	✓			
Presetup/Changeover Cart Part Scan		✓	✓	✓	✓	✓	✓
Presetup/Presetup Cart Operation		✓	✓	✓		✓	
Presetup/Prestup Part Unmount		✓	✓	✓	✓	✓	✓
Material Control/ Create	✓	✓	✓	✓			
Material Control/ Consume	✓	✓	✓	✓			

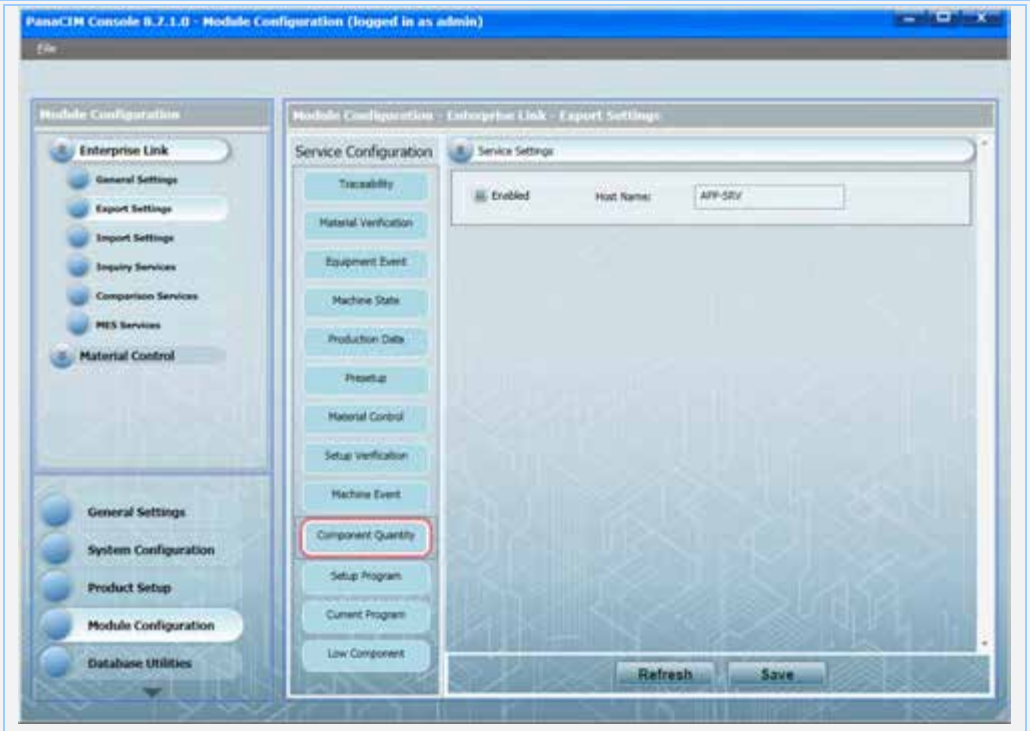
4.2 Settings

4.2.1 XML File

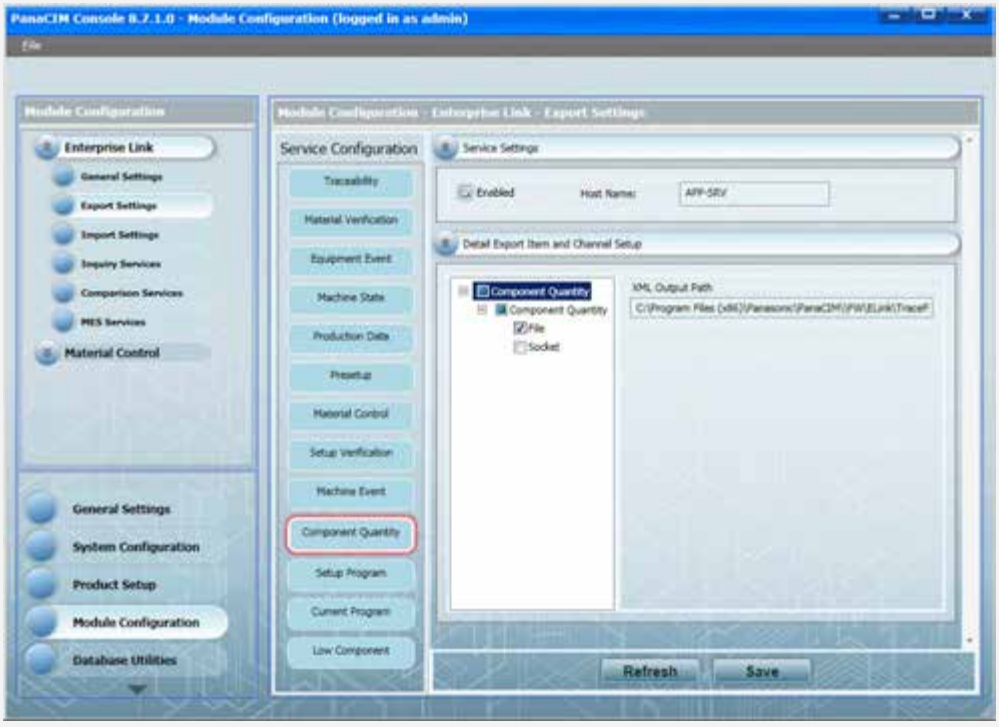
■ Configuring Export of Reel Quantity to XML file

PanaCIM can now export reel information for the reel mounted on a CM series (CM/DT) and NPM series (NPM/AM100) machine to an XML file whenever there is quantity change for the reels in a zone. It is customer's responsibility to manage these XML files. Important data that is exported via this functionality is the name of the current program, material being consumed by the current program, current remaining quantity of a material and feeder location of the material (Table number, slot and subslot). Material is identified by a Material id (i.e. Reel Barcode) in the XML file. It is to be noted that material needs to be consumed (i.e. part quantity in the reel need to change) for export to XML file.

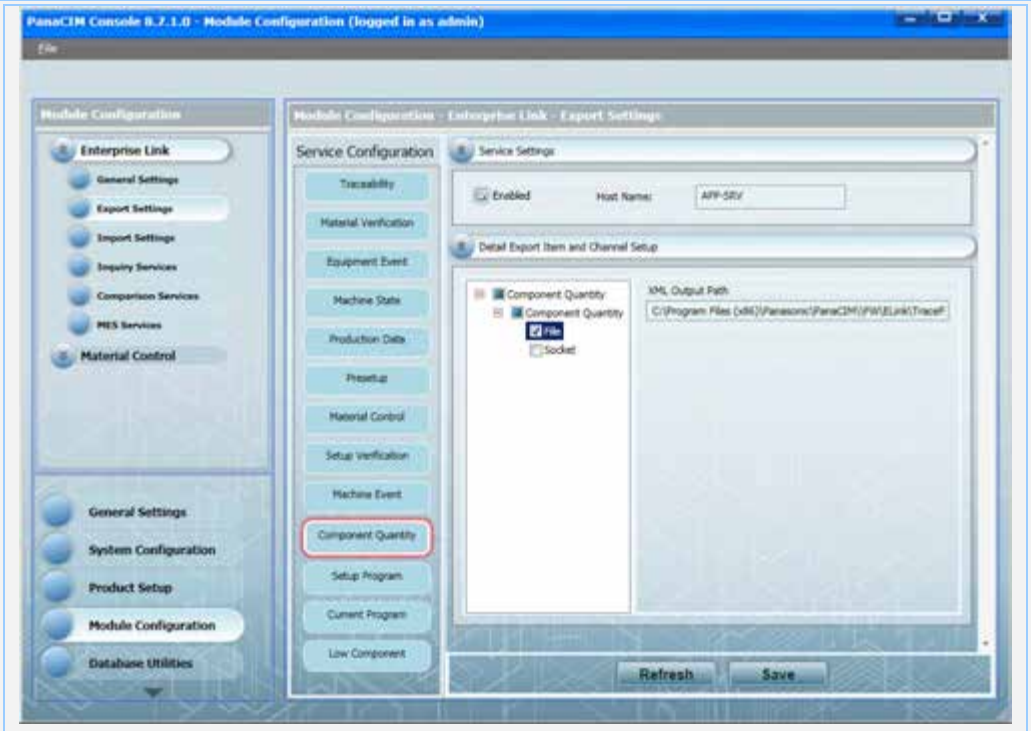
Step	Action
1	Launch PanaCIM Console and click **Export Settings** link within **Enterprise Link** plug-in after choosing **Module Configuration**. Following screen appears: 
2	Select **Component Quantity** from underneath **Service Configuration** heading. Following screen appears:



3	Check the **Enabled** checkbox under the **Service Settings** heading. Display changes on the right-hand side showing **Xml File** and **Socket** checkboxes under the **Component Quantity** checkbox as shown below:



4	Select the **Xml File** checkbox ensuring that is checked. A textbox appears showing default path where the XML file would be created.



It is to be noted that default path of XML file shown above cannot be changed. The XML file name is defined as: **(machine name) + Month + Day + Hours + minutes + Seconds + Milliseconds + .xml i.e. (machine name) + mm + DD + HH + MM + SS + MS + .xml)**.

5

Do one of the following:

Click the **Refresh** button to undo any configuration changes made and return to default settings existing for **Component Quantity** service.

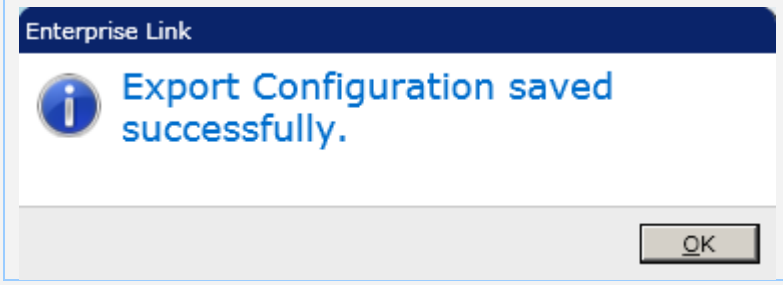
-OR-

Click the **Save** button to save the configuration changes made for **Component Quantity** service.

Note: After the **Save** button is clicked, it is not possible to undo the changes made by clicking the **Refresh** button.

6

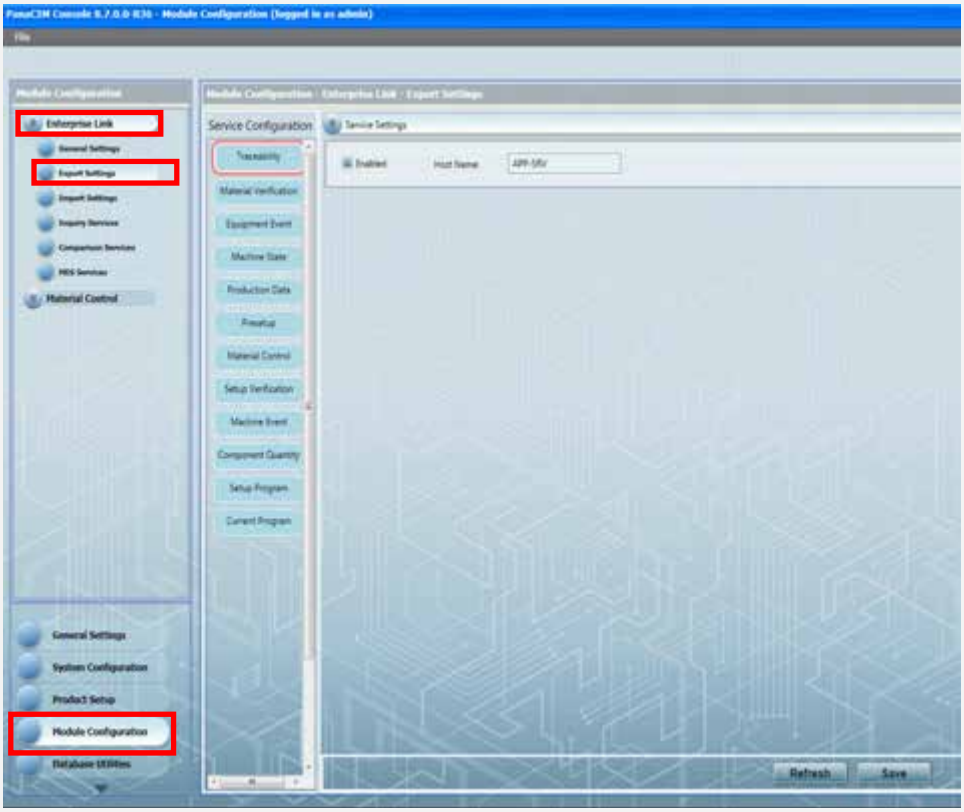
After the **Save** button is clicked, following message appears confirming that changes for **Component Quantity** service are saved:

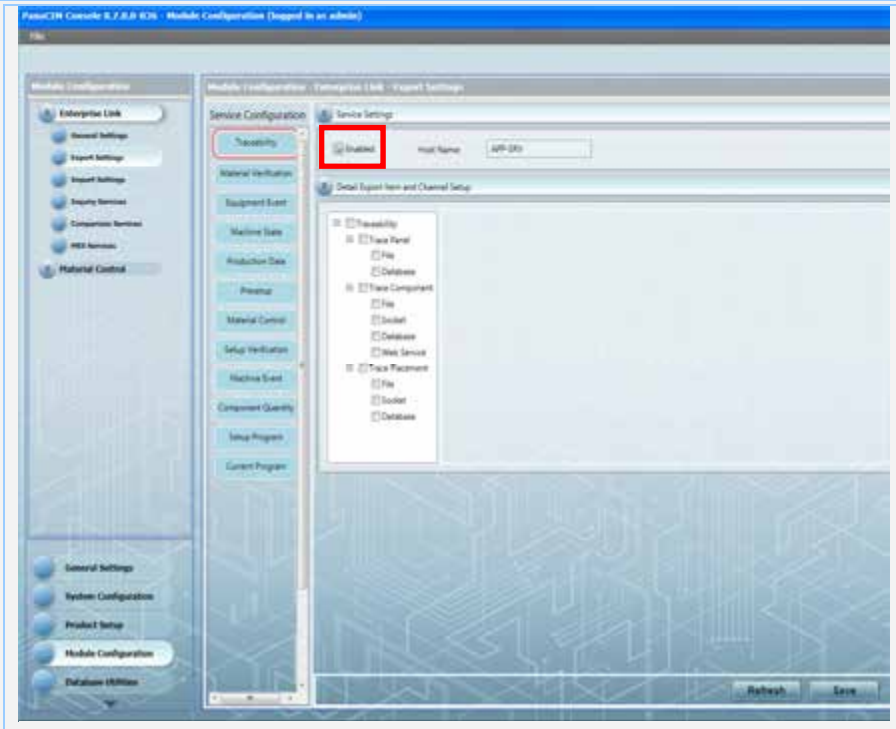


■ Exporting XML file of traceability

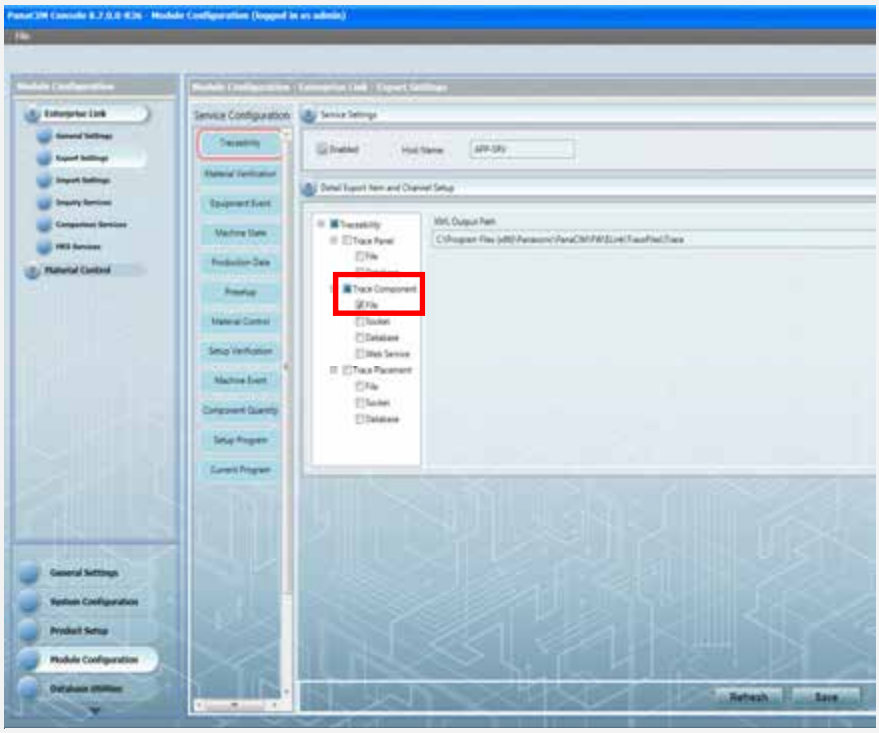
In PanaCIM, each time PCBs are produced for each stage by the CM-series machine (CM/DT) or the NPM-series machine (NPM/AM100), traceability can be output to the XML file as production information. Mount parts information on the produced PCBs is included in this XML file.
*XML files must be managed by a user. Output/accumulated XML files are not deleted automatically.

The following are the steps to configure the output settings.

Step	Action
1	Open PanaCIM Console > **Module Configuration** > **Enterprise Link** > **Export Settings.** Following screen appears. 
2	Mark the checkbox of **Traceability** in **Service Configuration** to enable. Following screen appears.

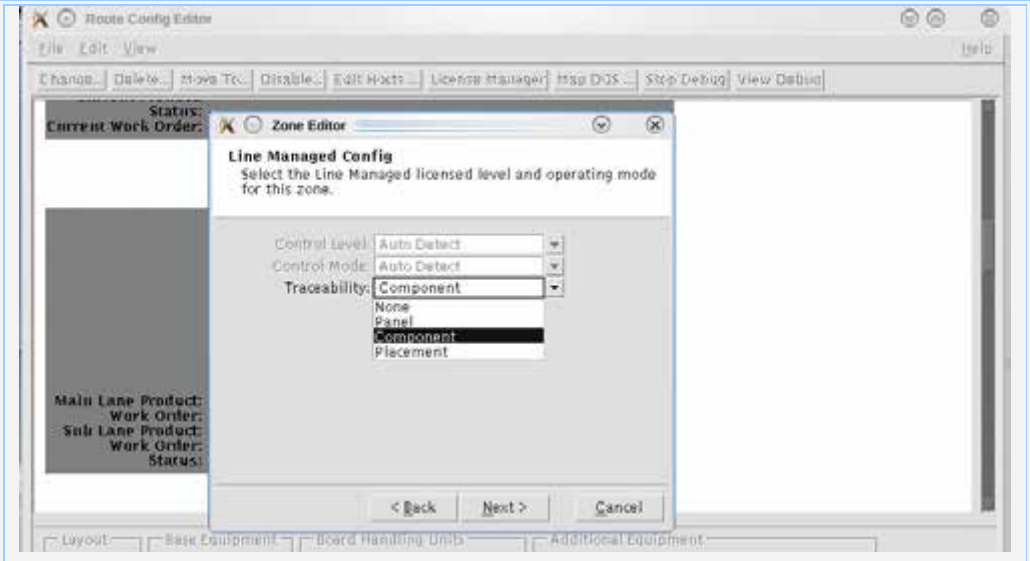


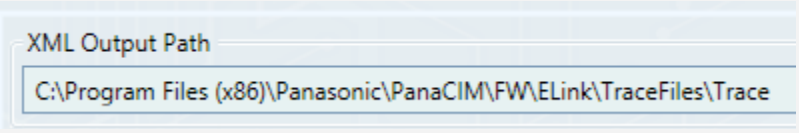
3	Mark the checkbox of any **File** in **Trace Panel, Trace Component or Trace Placement** to enable.



*The file you mark is based on Traceability under CIMC-> Route Config Editor-> Zone Editor->Line Managed Config.

・If it is “Panel”, mark the checkbox of File for Trace Panel.
・If it is “Component”, mark the checkbox of File for Trace Component.
・If it is “Placement”, mark the checkbox of File for Trace Placement.



4	Confirm **XML Output Path**. *XML files are output in this folder.  The XML file path cannot be changed. The XML filename is BARCODE+LINENO+STAGENO+CELLNAME+LANENO+STAGENO.xml.
5	Click on **Save**. 

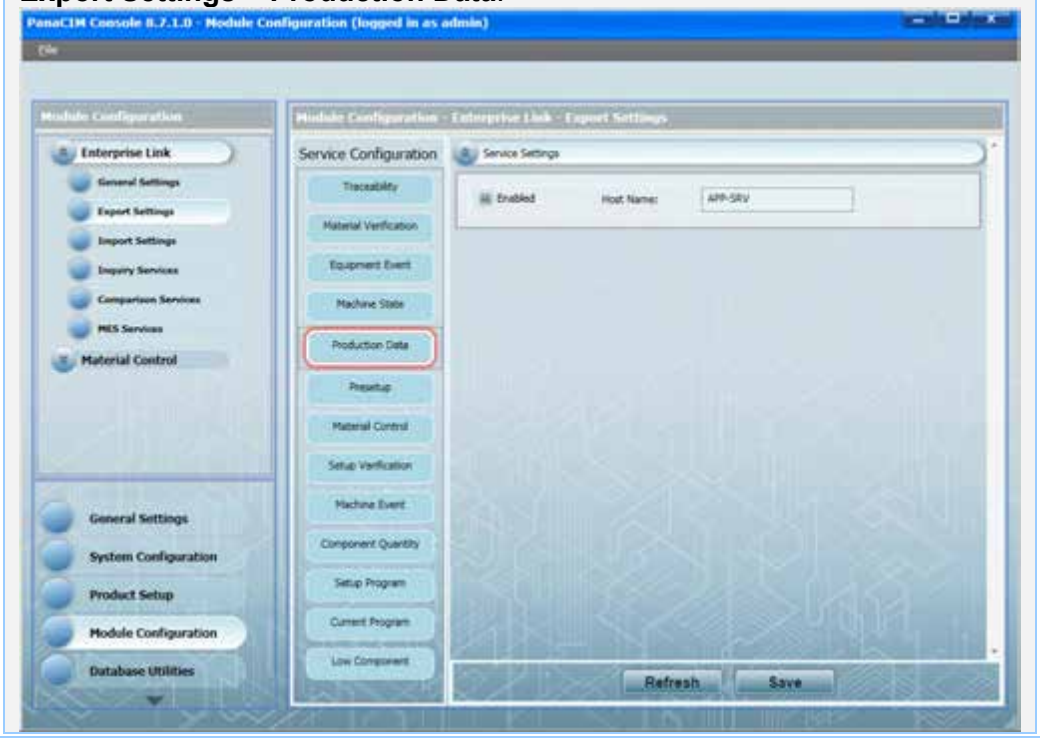
4.2.2 Database

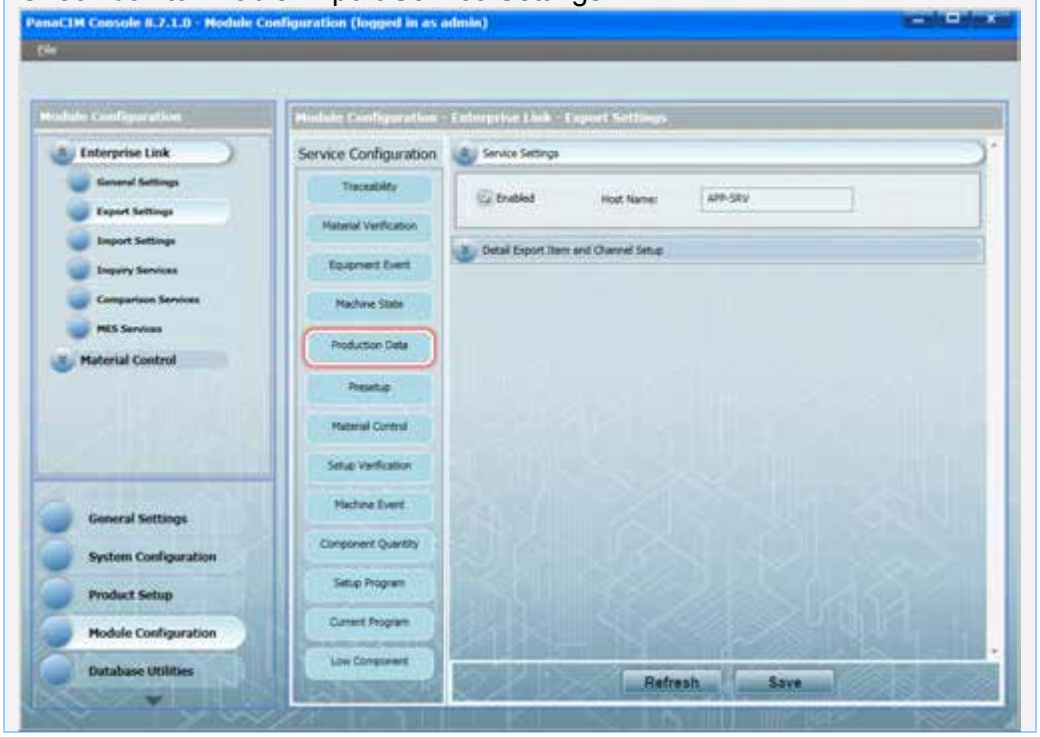
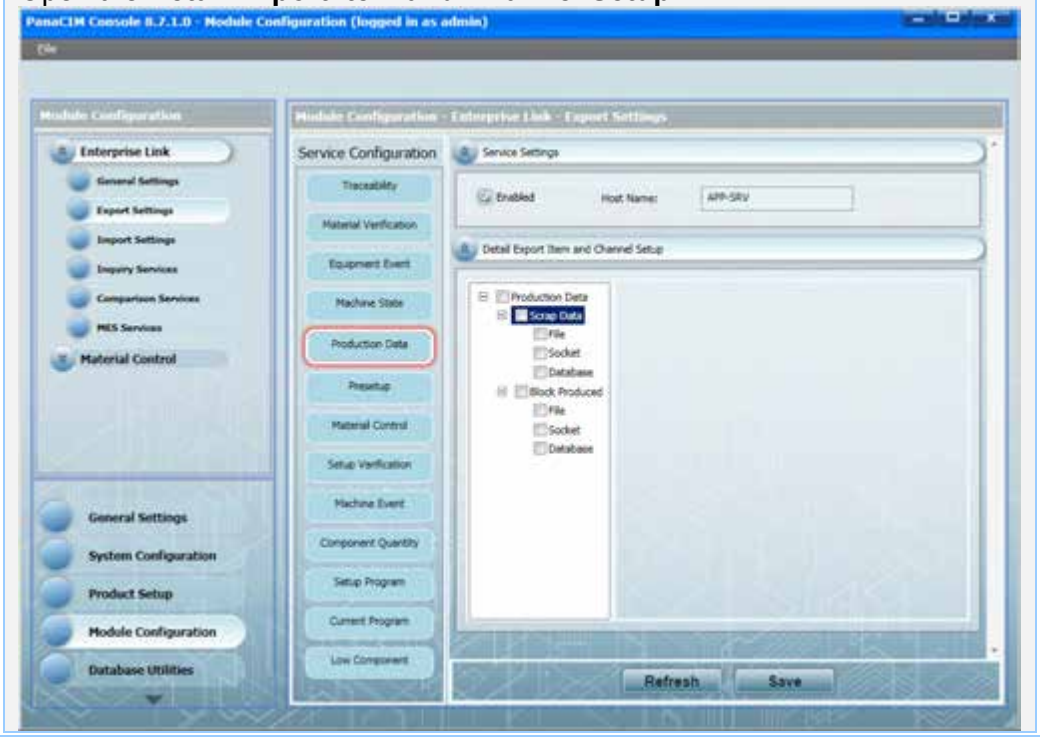
■ Assumptions

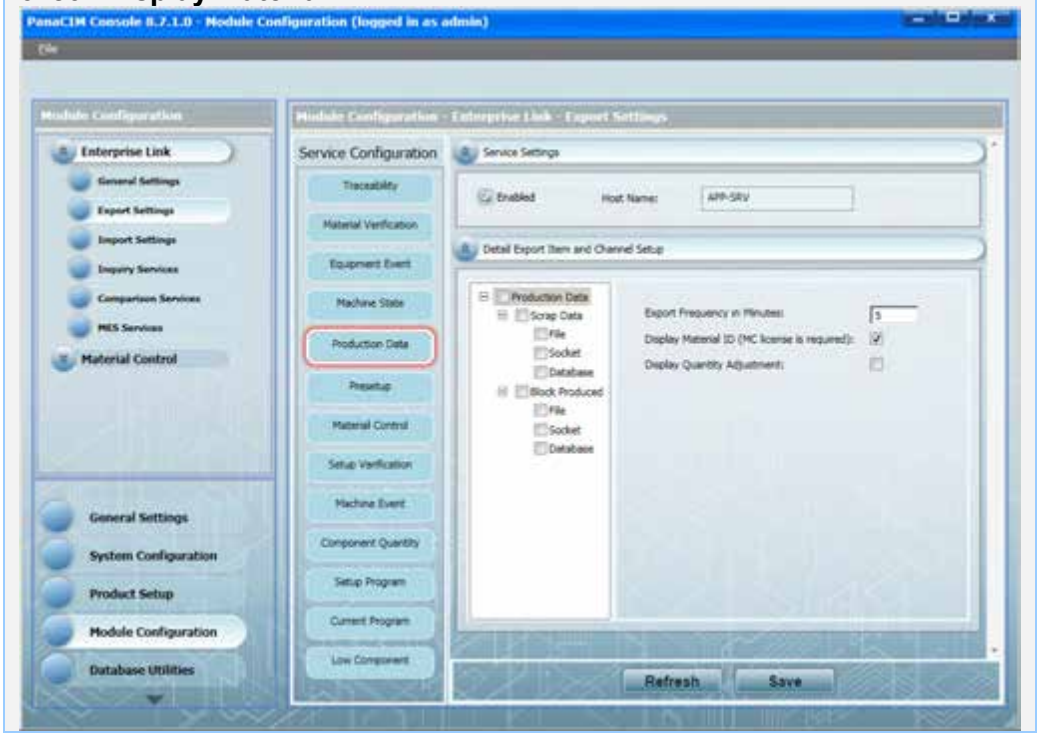
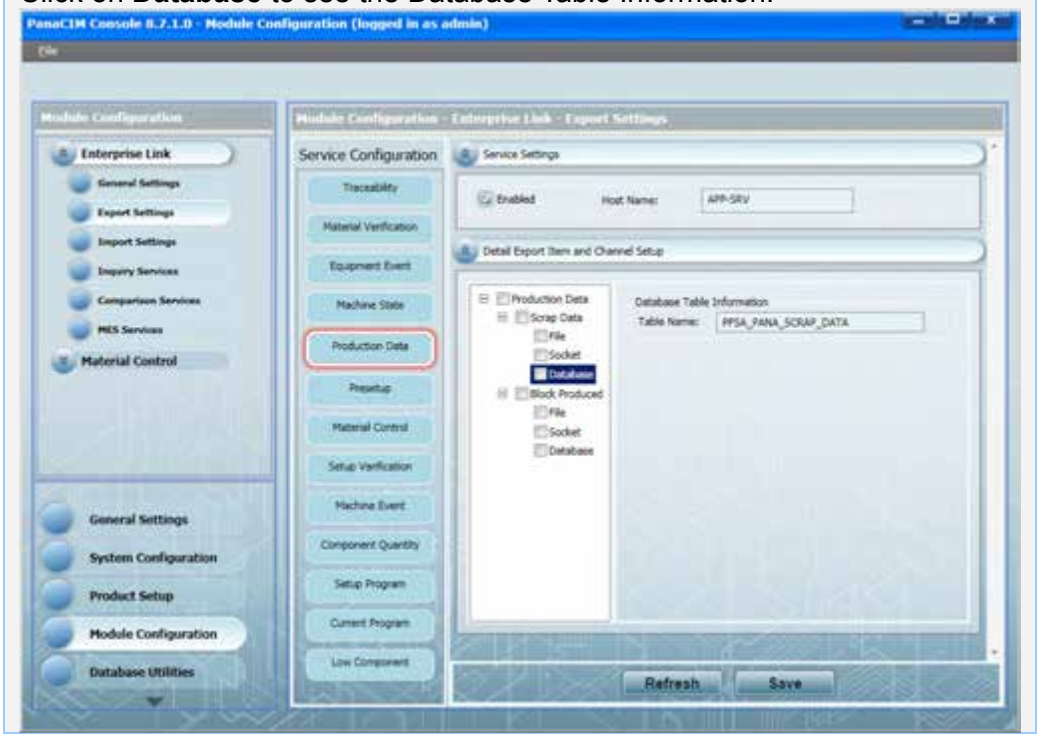
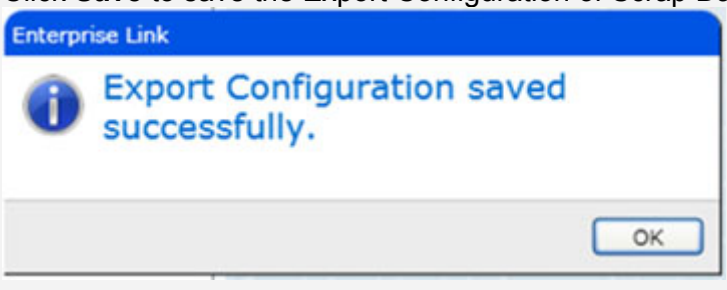
- Only support export scrap data to SQL server
- The Enterprise Link must have the permission to invoke the stored procedure in user's SQL server database.
- If Enterprise Link does not assign the permission to create table and stored procedure in user's SQL server database, manual setup to execute the script must be performed by the administrator.
- PanaCIM will provide a script to create a predefined table structure and store scrap data into the table in user's SQL database. It is user's responsibility to manage or maintain the data in the table.
- If the user does not want to use the predefined table, the user has to manually modify the stored procedure.
- If the data exporting to the customer's database failed due to any reason the user has to manually export the data through Enterprise Link web interface.

■ Configure Scrap Data Export

The following are the steps to configure Scrap Data Export:

Step	Action
1	Open PanaCIM Console > **Module Configuration** > **Enterprise Link** > **Export Settings** > **Production Data**. 

2	Check box to Enable Export Service Settings. 
3	Open the **Detail Export Item and Channel Setup**. 

4	Click on **Production Data** to set the **Export Frequency in Minutes** and check **Display Material ID**. 
5	Click on **Database** to see the Database Table Information. 
6	Click **Save** to save the Export Configuration of Scrap Data. 

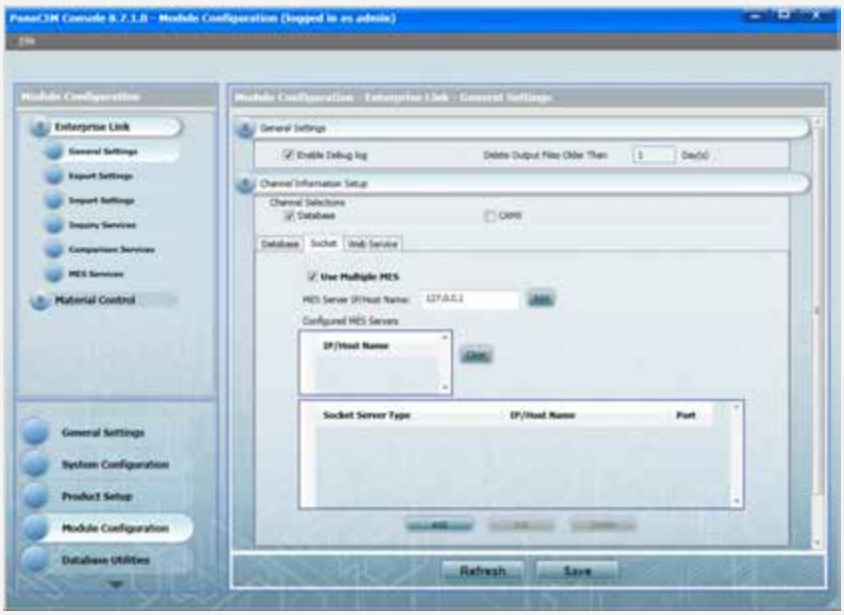
4.2.3 Socket Server Configuration

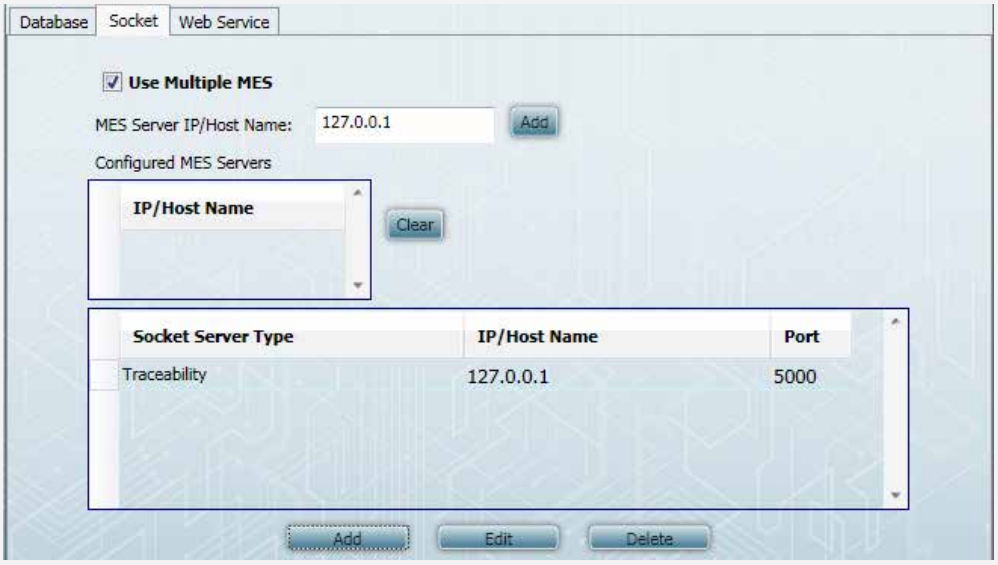
■ Configuration Enterprise Link to Send Data to MES

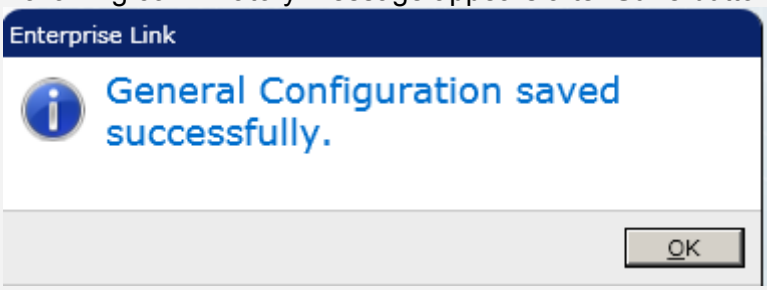
Enterprise Link can be configured to communicate with a third-party MES system to control manufacturing of PCBs. Manufacturing Execution System (MES) is a software system that aids or controls all aspects of the PCB manufacturing process. It is a master data source, which controls the quantity of a reel and the lifetime of MSD components. Production data for CM series (CM/DT) machines could be exchanged between Enterprise Link and MES.

● Socket Server Configuration

Socket Server configuration refers to the IP address and Port name of MES server to which PanaCIM sends data. It needs to be configured for each PanaCIM interface that needs to send data to MES.

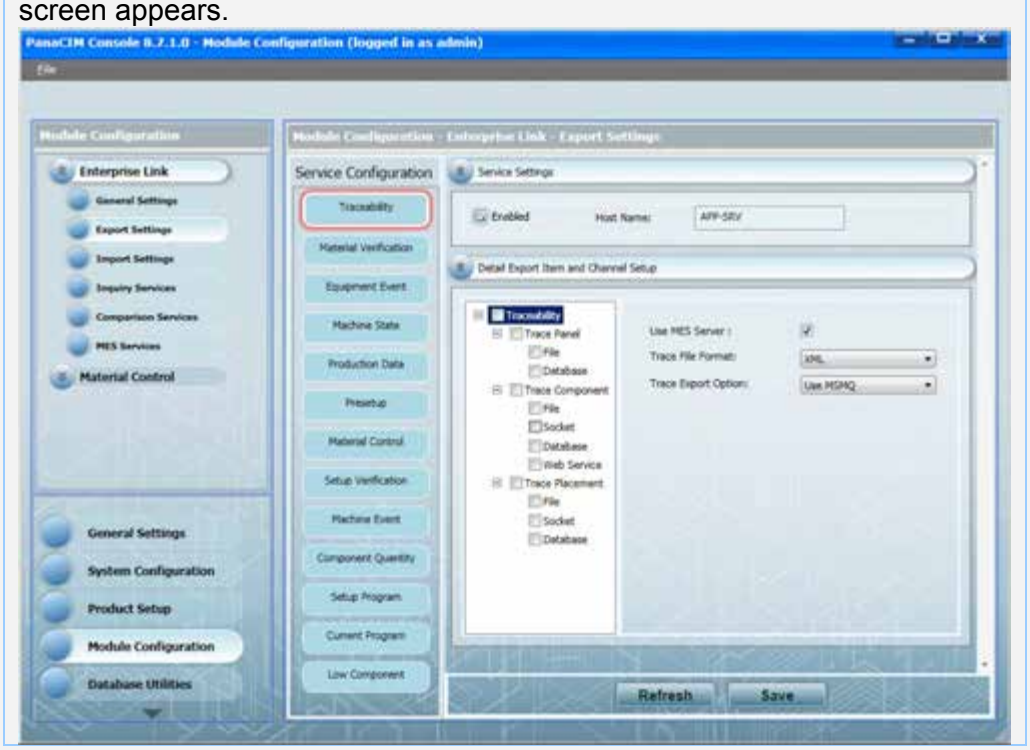
Step	Action
1	Launch PanaCIM console and select **General Settings** within **Enterprise Link** plug-in. Following screen appears: 
2	Ensure that the **Socket** tab is activated on the right-hand side as shown below: 

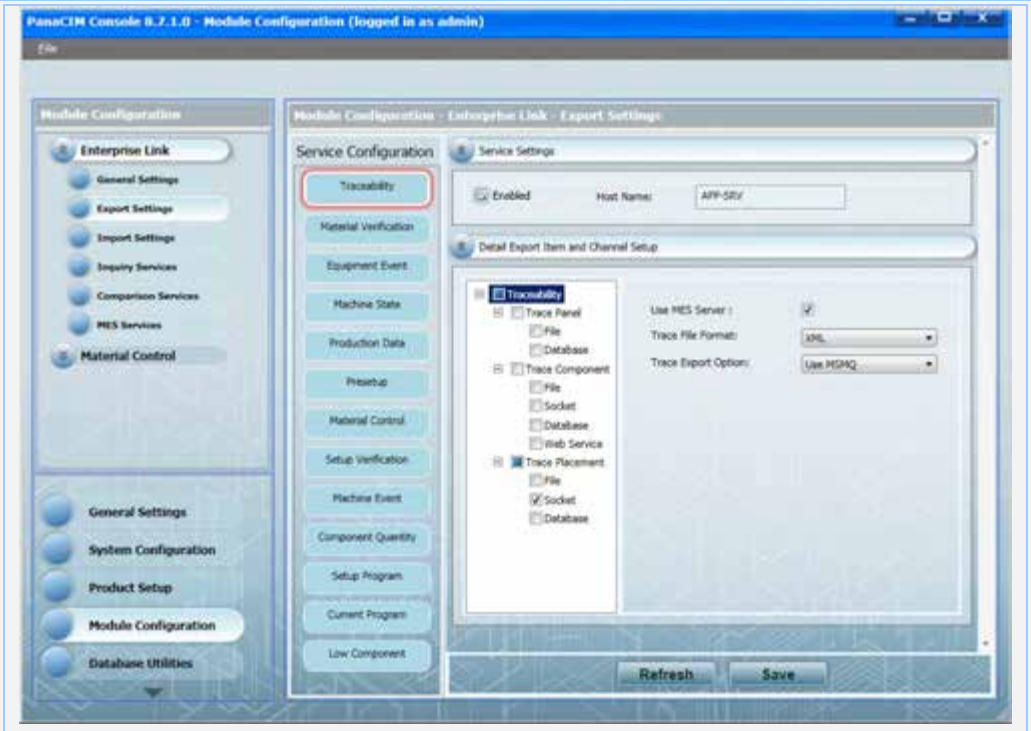
3	Click the **Add** button in the **Socket** tab. Following dialog appears:  Enter the IP address or the host name of the server running MES in the **IP/Host Name:** textbox. Enter a numerical identifier for a port in the **Port Number:** textbox.
4	Click the **OK** button in the **Add/Edit Socket Information** dialog after entering the **IP/Host Name** and **Port Number**. Following screen appears:  At this stage, following tasks could be carried out: • Click the **Add** button to add a new record for another PanaCIM Interface that needs to export its data to MES server. -OR- • Click the **Edit** button to edit details of an existing Socket Server connection. -OR- • Click the **Delete** button to delete an existing server socket connection. -OR- • Click the **Save** button to save changes made in the **Configuration** tab. It is to be noted that after clicking the **Save** button, it is not possible to revert back to earlier settings by clicking the **Refresh** button.

5	Following confirmatory message appears after **Save** button is clicked. 

- **Enabling MES for a PanaCIM Interface**

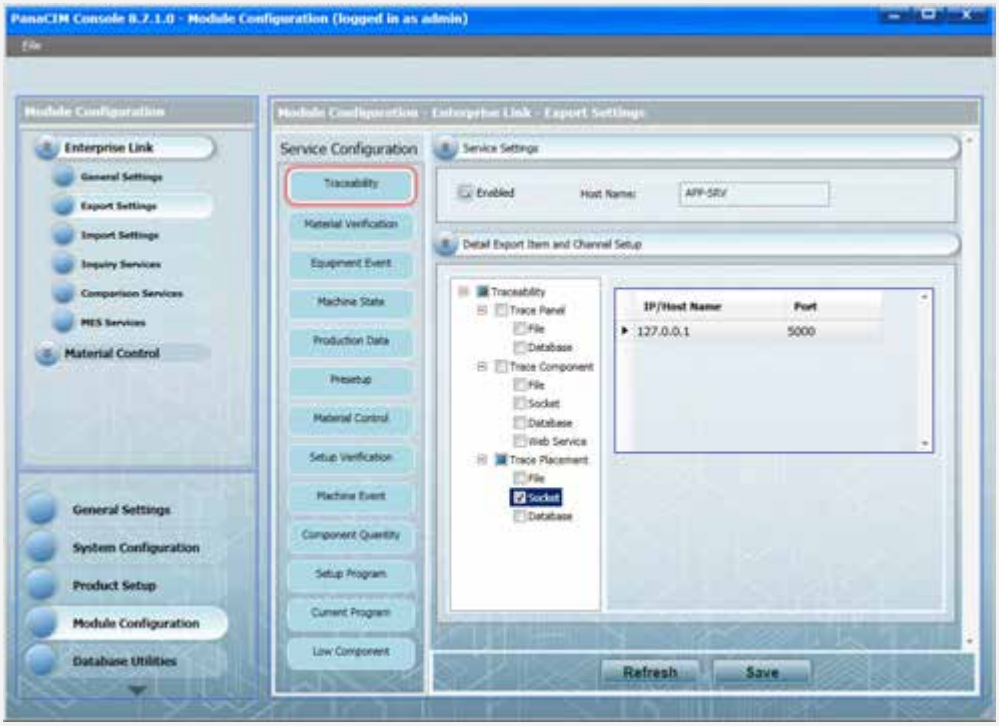
MES needs to be enabled for PanaCIM Interface for which IP address and Port name of MES server was specified using procedure outlined in section above. Refer to the action table below.

Step	Action
1	Activate **Export Settings** tab within **Enterprise Link** plug-in. Following screen appears. 
2	Under the **Service Configuration** heading, select the PanaCIM interface that needs to be enabled for exchanging data with MES. For example, if PanaCIM needs to send/receive messages related to Panel Traceability, select **Traceability** under **Service Configuration** and then select **Traceability** under the **Detail Export Item and Channel Setup** heading.



Check **Use MES Server** to enable exchange of traceability messages between PanaCIM and MES.

At any time, IP address/Port name of MES server can be viewed by clicking socket as shown below.

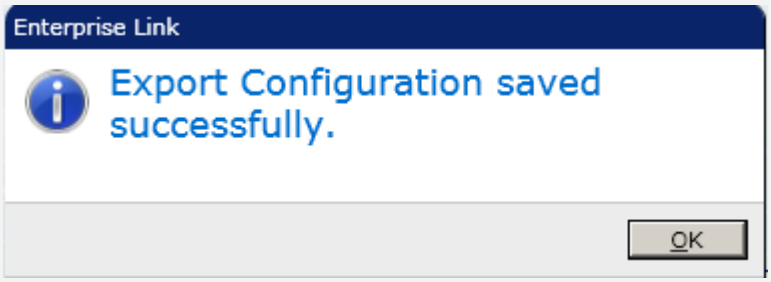


Note: Placement Trace Data can be exported to third party MES server in addition to component trace.

3	Do one of the following: • Click the **Refresh** button to return to default settings that existed prior to making any changes.

-OR-

- Click the **Save** button to save changes. After clicking the save button, following confirmatory message appears:



4 Repeat steps 1-3 for each PanaCIM interface that needs to be configured to use MES server.
*Interfaces that can be supported (tested) by current PanaCIM are as follows.

- Traceability
  - Trace Placement
- Material Verification
  - Part Scan
  - Part Exhaust
  - Splicing Scan
- Presetup
  - Changeover Cart Part Scan
  - Presetup Cart Operation
  - Presetup Part Unmount
- Setup Verification
  - Setup Check
- Material Control
  - Material Removal
- Machine Event
  - PCB End
- Component Quantity
  - Components Quantity per Route
- Current Program
  - Current Program

5. FUNCTION - ENTERPRISE LINK UTILITY

This chapter explains function-Enterprise Link Utility.

5.1 Overview

The following PanaCIM-EE Enterprise Link Utilities are explained.

- MCID Management Utility
- Message Log Utility

5.2 Settings

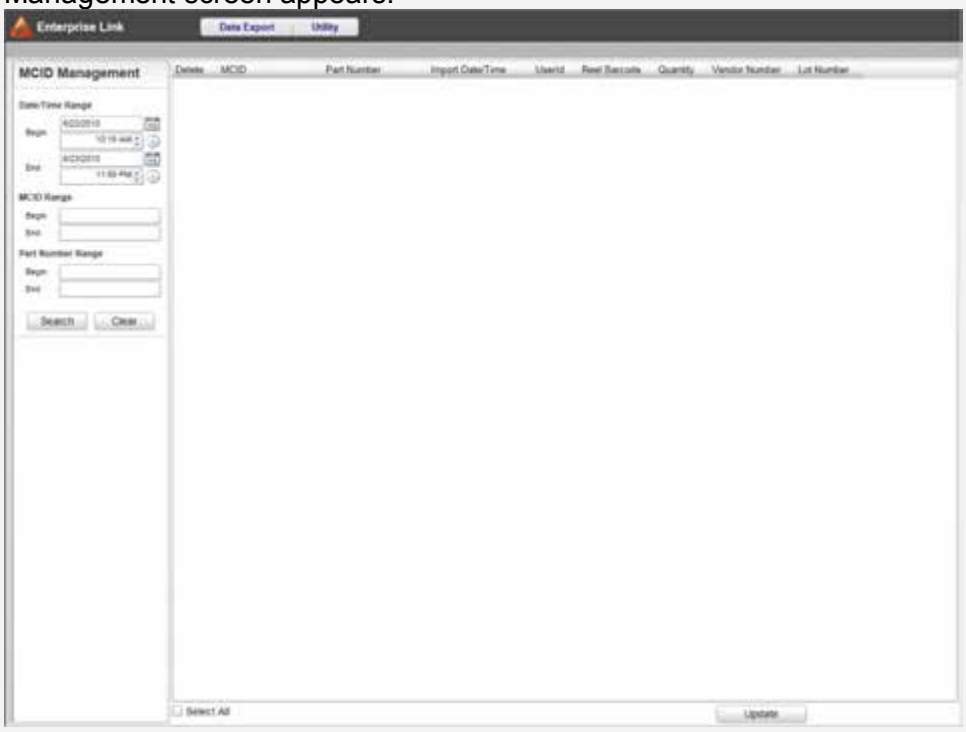
Please refer to 5.3 Operation.

5.3 Operation

5.3.1 MCID Management Utility

Use the MCID Management Utility to delete/update records in the database table.

Refer to the action table below to modify records using the MCID Management Utility.

Step	Action
1	Select the **Enterprise Link** icon and select the **Utilities** link.
2	Select the **Utility** menu item and click **MCID Management**. The MCID Management screen appears. 
3	a. Define the **Date/Time Range** when the User may have File Processed the File, or enter the **MCID Range** or **Part Number Range**. The List of MCID, Part Number, and The Import Date/Time will appear in the right hand side of the screen. *Optional*: Modify the Part Number if desired and click the Update button. Delete a record by selecting the checkbox and clicking the Update button. b. Click **Search**.

5.3.2 Message Log Utility

Refer to the action table below to access the Message Log Utility to view Enterprise Link messages for a specified time period.

Step	Action
1	Select the **Enterprise Link** icon and click the **Utilities** link.
2	Click **Utility** and select **Message Log**. 
3	Enter a **Date and Time Range** in the appropriate fields and click **Search**.

6. APPENDIX

6.1 Export Formatting Examples

6.1.1 Traceability

■ Socket Formatting

A sample format for a panel trace is as follows:

```
000342(BID0024)WBIIB0000100000(BNR0043)DIMA;DIMALOT;DIMAFIL
E;DIMAFILE;0.1(BNE0023)CM402 KXF-4Z4C(BZE0028)10.09.2004-
14:28:05(BLS0010)3(BLP0014)30007(BLT0010)3(BLF0010)L(BLE001
2)***(BCB0019)AZ08054501(BCG0034)AZ08054501 MYVENDOR
12345(BLS0010)4(BLP0014)40008(BLT0010)4(BLF0010)L(BLE0012)*
**(BCB0019)AZ08054502(BCG0034)AZ08054502 MYVENDOR 12345
```

Reference the following table for component descriptions.

Component	Description
00342	Total length of the message sent to the socket.
BID	Barcode
0024	Total length of the barcode (including parenthesis)
BNR	Placement Program (MAI file name)
BNE	Equipment Identifier Lane Stage. This is the name of the machine as defined in the Factory Explorer Application, suffixed by lane and stage format.
BZE	This is the TIME the panel left the machine in dd.mm.yyyy-hh:mm:ss format
BLS	Table number
BLP	Slot
BLT	Also the table number
BLF	The side of the feeder. If you have a double feeder, thin-type single feeder, or automatic loading feeder (spliceless feeder), this will read **R** or **L**. Single feeders always read **L**.
BLE	Feeder ID
BCB	Part number
BCG	Part number followed by the Vendor Lot Number.

■ XML Schema

```
PANEL>
BOARD LINENO="1013" CELLNO="1049" CELLNAME="CM401-1" LANENO="0" STAGENO="1" PRODUCTNAME="SIIX-
 001" SETUPNAME="S-001" PRODID="AA" BARCODE="AA0011" SIDE="Top" PROGID="SI-001:SIIX-001:SIIX-001:"
 TIMEOUT="12/08/2009 03:49:43 PM">
FEEDER FEEDERNO="108" TABLENO="2" SLOT="20027" SUBSLOT="1" PARTNO="ERZVA5D471" LOTNO="7021"
 EXPECTEDPART="" VENDORNO="1207" FEEDERID="FA0210AYA001333" QUANTITY="2000" TIMEOFF="12/8/2009
 3:49:43 PM" OPERATORNAME="" USERDATA="" />
 </BOARD>
 </PANEL>
```

6.1.2 Material Verification

■ XML Schema

Note: The name is defined as:

P + Month + Day + Hour + minutes + Seconds + Milliseconds + .xml

(P + mm + DD + HH + MM + SS + MS + .xml)

```
<?xml version="1.0" encoding="utf-8" ?>
<PARTSCAN><FEEDERDATA HEADID="1127" LINENO="1022" LINENAME="001" ZONENO="1069"
  ZONENAME="Zone-41" CARTID="" CELLNO="1127" CELLNAME="Zone-41-Machine-1"
  LANENO="0" STAGENO="1" Z="12" SLOT="10006" SUBSLOT="1" PARTNO="T1608-04072009-
  1001L" EXPECTEDPART="" LOTNO="P00LL" VENDORNO="P00LL"
  FEEDERID="FA0210AJA453139" QUANTITY="416" PARTTIMEON="07/07/2010 11:45:40"
  PARTTIMEOFF="07/07/2010 11:59:50" OPERATORNAME="444" OPERATION="Part Scan"
  USERDATA="" REELBAR="FJ07072010-001" MCID="" PARTCLASS="111" COMPARISONID=""
  OVERRIDE_REASON="" ></FEEDERDATA></PARTSCAN>
```

6.1.3 Machine State

■ XML Schema

Note: TIMESTAMP format is “YYYYMMDDHHMMSSSSS”

```
<?xml version="1.0" encoding="UTF-8" ?>
 - <equipment xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:noNamespaceSchemaLocation="equipment_state-1.1.xsd">
 <state Name="RUNNIGN" Equipment="CM402 KXF-4Z4C" StageNo="0" Timestamp="2008-07-
  17T15:26:59+05:30" />
 </equipment>
```

6.1.4 PreSetup

■ XML Schema

```
<?xml version="1.0" encoding="utf-8" ?>
<PARTSCAN>< FEEDERDATA HEADID="" LINENO="1001" LINENAME ="ROUTE1"
  ZONENO="1001" ZONENAME="ZONE2" CARTID ="12345" CELLNO=""
  CELLNAME="" LANENO="" STAGENO="" Z="" SLOT="17" SUBSLOT="1"
  PARTNO="0402" EXPECTEDPART="" LOTNO="3" VENDORNO="3"
  FEEDERID="FAXXXXX" QUANTITY="3" PARTTIMEON="09/03/2009 05:32:40"
  PARTTIMEOFF="" OPERATORNAME="111" OPERATION="PreSetup Scan"
  USERDATA="3" REELBAR="R001" PartAlias="TCTC02X800098"
  >< /FEEDERDATA></PARTSCAN>
```

6.1.5 Production

■ XML Schema

```
<?xml version="1.0" encoding="utf-8" ?>
- <PRODUCTIONDATA>
    <LINE NAME="AAA4" />
    <CELL NAME="CM401-1" />
    <STARTTIME VALUE="2010-01-21T16:13:48-06:00" />
    <ENDTIME VALUE="2010-01-21T16:33:19-06:00" />
  - <DATALIST>
      <DATA LANENO="1" STAGENO="1" PRODUCTNAME="LC-003" BLOCKPRODUCED="6" BOARDPRODUCED="1" />
    </DATALIST>
  </PRODUCTIONDATA>
```

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